

When Protocol States Mislead: Evidence-Gated Admission and Online Feedback Calibration for LLM-Assisted Stateful Fuzzing

Ketang Chen^a, Xiaolei Ren^{a,*}

^aMacau University of Science and Technology, Avenida Wai Long, Taipa, Macau, China

Abstract

Response-derived states offer inexpensive feedback for stateful protocol fuzzing, but state novelty is an imperfect proxy for code progress. Requests generated by large language models (LLMs) can amplify this mismatch when unproductive proposals receive durable mutation resources. We present LoopFuzz, an evidence-calibrated controller design combining execute-before-promote admission with online state-productivity calibration. Bounded trials separate structural validity, response acceptability, and reachability from code and inferred protocol state machine (IPSM) novelty. Code-supported candidates qualify for durable admission; state-only candidates receive provisional validation and expire without subsequent code evidence. Discounted Beta-Bernoulli updates and Thompson sampling combine estimated code productivity with frontier exploration, without claiming to recover semantic protocol states. A retrospective analysis of 628 experimental records across nine targets provides descriptive coverage, cost, and mechanism measurements. Observed code-branch and IPSM state-edge endpoints do not consistently reproduce the hypothesized mismatch across motivating targets. Four vulnerability case studies add replay evidence; controlled rediscovery benefits remain unmeasured. A five-arm evaluation protocol specifies matched model configurations, intervention policies, and resource ceilings to isolate admission and calibration. Heterogeneous observation horizons and incomplete lifecycle evidence restrict current findings to descriptive results.

Keywords: Stateful protocol fuzzing, Large language models, Response-derived state feedback, Evidence-gated admission, Online productivity calibration

1. Introduction

Stateful protocol fuzzing must discover useful interaction sequences as well as useful message contents. A server can accept a request only after earlier exchanges establish a session, authenticate a client, or create a resource. AFLNet uses responses to construct an inferred protocol state machine (IPSM), making protocol-visible feedback available without a target-specific internal-state extractor [1]. This deployment advantage comes with an abstraction boundary: a response code or sequence is evidence about observable behavior, rather than an identifier of the server's complete semantic state.

StateAFL already identifies limitations of response-derived state abstractions and uses memory observations to infer server states [2]. The fact that responses can misrepresent internal progress is therefore established background. Our question is how a fuzzer retaining a response-visible abstraction should decide, during a campaign, whether a particular state signal is useful enough to justify further exploration. A new IPSM state edge can accompany a previously unseen code branch, reproduce an already explored rejection path, or distinguish responses after the measured code surface has saturated. These outcomes should not receive identical evidential weight.

Assistance from large language models (LLMs) makes this distinction consequential for queue management. ChatAFL demonstrates that LLMs can derive protocol structures and propose interaction sequences [3]. A generated request is nevertheless a hypothesis about both message structure and live session conditions. If a proposal is retained indefinitely because it produces a new response-derived state, the fuzzer may repeatedly mutate an input whose apparent novelty has little downstream code value. Durable queue membership represents a continuing allocation of execution resources.

Three target families motivate the investigation. Forked-daapd exposes DAAP/DACP behavior through HTTP status responses, so response distinctions may overstate useful code progress. Lighttpd1 can execute different code paths under the same HTTP status, so response novelty may understate code progress. LightFTP motivates examining continuing IPSM growth after code-coverage saturation. These hypotheses motivate the evaluation; they are not assumed to hold in the present observations. Section 2 separates them from the available measurements.

LoopFuzz addresses this problem through evidence-gated admission and online state-productivity calibration. It first checks a proposal's structure, then executes a bounded trial and separates response acceptability, state reachability, code novelty, and IPSM novelty. Code-supported candidates can enter the durable queue. State-only candidates receive a provisional validation

*Corresponding author: Xiaolei Ren.

Email addresses: 3250006913@student.must.edu.mo (Ketang Chen), x1ren@must.edu.mo (Xiaolei Ren)

budget and must earn durable promotion through subsequent code evidence. At the scheduling layer, a discounted productivity estimate conditions investment on the observed relationship between selected response states and later code progress.

This work makes three contributions, with their current evidence status stated explicitly.

1. **Evidence-gated queue admission.** We specify an execute-before-promote controller that distinguishes reject, provisional, and durable outcomes. It separates state novelty from code evidence and makes the lifetime and descendant cost of provisional entries traceable.
2. **Online calibration of state productivity.** We formulate response-state utility as the probability of code progress in a bounded state-selection episode. Discounted Beta-Bernoulli updates and Thompson sampling instantiate the controller; these established estimation and selection methods are not claimed as new algorithms.
3. **A controlled empirical framework with partial evidence.** We define candidate-level and episode-level measurements, five experimental arms, complete trial accounting, and information-controlled rediscovery on vulnerable/patch-paired targets. We analyze the available experimental evidence, report four vulnerability replay cases with explicit limitations, and identify the observations needed for causal admission and calibration claims.

The available evidence does not justify describing the third contribution as a completed five-arm evaluation. The distinction between the proposed method and its empirical support is maintained throughout the evaluation.

2. Background and Motivation

2.1. Response-visible state and independent code feedback

Let $\mathcal{M}_t = (Q_t, \widehat{\Delta}_t, q_0)$ be the IPSM observed by time t , where Q_t contains response-derived state identifiers, $\widehat{\Delta}_t \subseteq Q_t \times Q_t$ contains observed IPSM state edges, and q_0 denotes the initial connection state. The abstraction can alias different internal states and separate executions that reach the same code. It can also depend on history, asynchronous replies, and reset behavior. It is not assumed deterministic or semantically complete.

Let $\mathcal{B}_t$ be the covered code-branch set under a fixed coverage instrumentation and replay procedure. An IPSM state-edge increment changes $\widehat{\Delta}_t$; a code-branch increment changes $\mathcal{B}_t$. These objects have distinct units. An AFL bitmap observation must also distinguish a newly encountered code edge from a hit-count bucket change. Neither bitmap slots nor response-state identifiers are interchangeable with source-level code branches.

2.2. Three falsifiable mismatch cases

Table 1 defines the three motivating cases. Figure 1 summarizes the observed A/D endpoints as mean $\pm$ sample standard deviation (SD), with actual run counts n and nominal $N = 10$.

Establishing a mismatch mechanism requires comparable executable versions, a shared observation horizon, and execution-level or time-resolved evidence. Endpoint differences alone do not establish which state caused a difference. Contradictory observations are retained when assessing the motivating hypotheses.

The desired controller must accommodate both overestimation and underestimation. Rejecting every state-only candidate could lose useful bridges; promoting every such candidate would treat an imperfect proxy as sufficient evidence for permanent resource allocation. A provisional queue expresses the intermediate conclusion that an input deserves limited descendant validation while its longer-term value remains unproven.

The scope is textual or semi-textual request-response protocols with usable parsers, response extractors, and code feedback. Response-visible state extraction avoids a target-specific internal-state adapter, but the complete controller remains grey-box because productivity requires code feedback. It is not a pure black-box system. Encrypted or opaque binary protocols require suitable construction, response, and code-observation adapters before evaluation.

3. Related Work and Novelty Boundary

Stateful greybox fuzzing studies how to expose, infer, and schedule protocol state [1, 2, 4, 5]. AFLNet supplies a response-derived IPSM, whereas StateAFL and NSFuzz use different internal observations. The subsequent AFLNet study revisits coverage-guided protocol fuzzing [6], while greybox protocol-state learning investigates how internal observations inform inferred models [7]. LoopFuzz retains the response-visible interface and studies its instrumental reliability under an independent code-progress criterion. It does not claim to originate state inference or to first identify misleading responses.

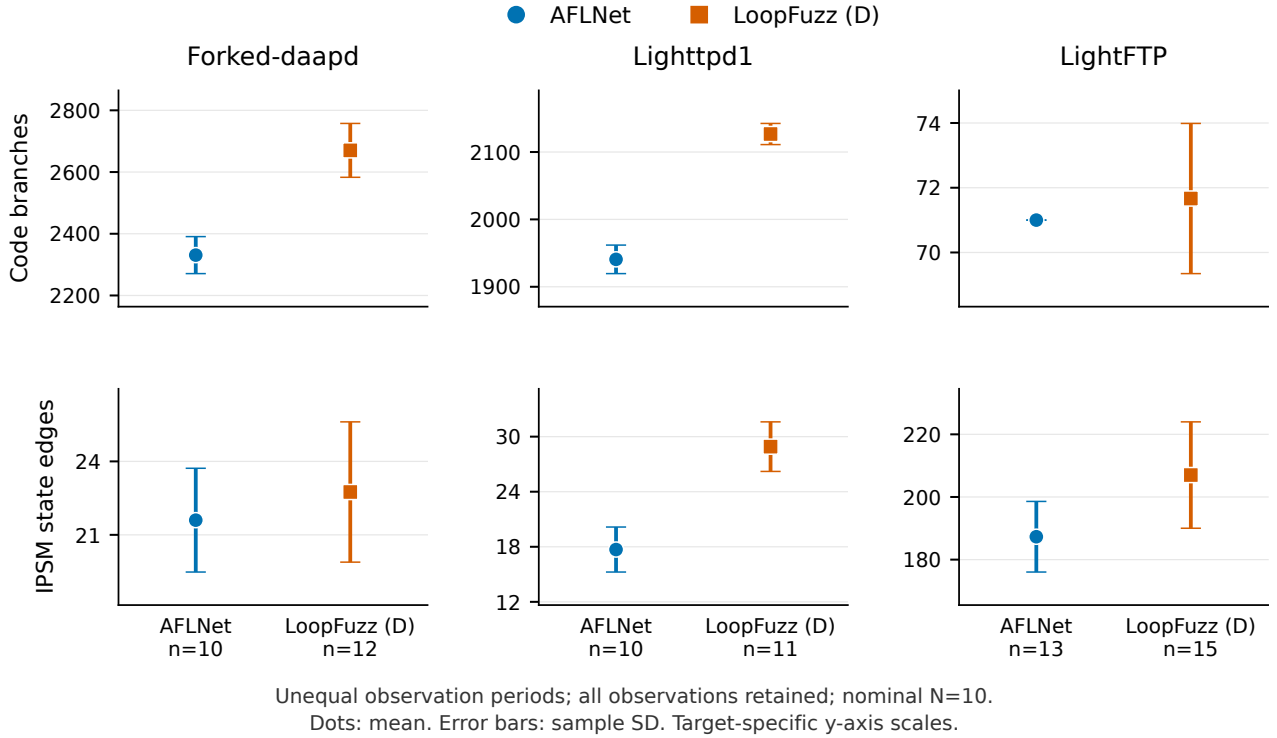
Execution control and feedback interpretation address complementary problems. Nyx-Net uses incremental snapshots for network fuzzing [8]; SnapFuzz targets high-throughput execution of network applications [9]; fault-injection-based network fuzzing provides another approach to exercising protocol interactions [10]. BLuEMan simulates interactions between actual Bluetooth Low Energy (BLE) stacks [11]. DistFuzz combines events, faults, and timing with message-sequence feedback and symmetry-based pruning for distributed systems [12]. These studies motivate treating execution conditions and observable feedback as separate design choices. Their protocol domains and execution mechanisms are not evidence that LoopFuzz generalizes beyond its evaluated servers.

Structured generation reduces the difficulty of producing admissible inputs. Gramatron and CarpetFuzz exploit grammars and documented constraints [13, 14]. FormatFuzzer derives parsers, mutators, and generators from binary-format specifications [15]; Fuzztruction instead injects faults into a generating program to retain implicit format knowledge [16]. MINER uses valid sequence templates and learned request parameters to guide later requests [17]. These approaches concern how candidates are constructed. Our admission rule asks a subsequent question:

Table 1: Motivating hypotheses and current evidence.

Target	Hypothesis	Required observation	Current interpretation
Forked-daapd	Overestimation: IPSM growth without downstream code gain.	Growing IPSM structure with low episode code reward.	Prior endpoint code regression not reproduced.
Lighttpd1	Underestimation: code progress within shallow response states.	Code-branch gain with little IPSM gain, linked to executions.	Both endpoints increase; hidden productivity remains unverified.
LightFTP	Saturation: IPSM novelty after code progress plateaus.	Aligned code/IPSM trajectories across saturation.	Near-equal endpoints; matched-version trajectories still needed.

Observed code and IPSM endpoints

Figure 1: A/D code and IPSM endpoints (mean $\pm$ SD; actual n shown).

whether a candidate has enough independently observed code value to acquire durable queue resources.

LLM-assisted fuzzing extends generation to protocol interactions, device inputs, and programs [3, 18, 19]. FuzzGPT studies unusual programs for deep-learning libraries [20], and WhiteFox uses compiler source information to guide optimization-triggering tests [21]. ProphetFuzz predicts high-risk option combinations from documentation [22]. Recent studies also address LLM-assisted hybrid fuzzing [23] and predicate-guided synthesis for directed testing [24]. These varied generation objectives reinforce the distinction between a model’s proposal and evidence obtained by executing it. The comparison specified here preserves ChatAFL’s scheduling lineage. Admission and calibration require separate controlled contrasts.

Adaptive resource allocation is established in fuzzing. The Bandit’s States models protocol-state selection as a bandit problem [25]; Thompson-sampling seed scheduling [26] and state-significance-guided protocol fuzzing [27] further delimit the

contribution. FOX formulates coverage-guided fuzzing as online stochastic control [28], while FISHFUZZ dynamically prioritizes targets and seeds using distance information [29]. WingMuzz studies two-dimensional scheduling for blackbox protocol testing [30]. Thus neither adaptive selection nor a productivity score is novel by itself. The distinction here is the combination of a response-derived proxy, an independent code-progress target, provisional and durable LLM-candidate admission, and traceable episode-level evidence.

Protocol correctness also requires an oracle beyond apparent acceptance. ProtocolGuard combines specification-derived checks with dynamic verification of non-compliance [31]; BSFuzzer uses context-aware semantic testing for BLE logic flaws [32]; MerCuriuzz targets logical vulnerabilities in QUIC implementations [33]. These studies examine semantic violations, whereas the proposed calibration reward targets code productivity. A response-state transition or positive code reward therefore does not certify security impact. Magma’s

Table 2: Prior mechanisms and novelty boundaries.

Prior work	Established focus	Present boundary
AFLNet; StateAFL; NSFuzz	Response-derived or internal-state feedback.	Calibrate the response proxy; no semantic-state recovery.
ChatAFL; LLM synthesis	Protocol-informed message/sequence generation.	Execute-before-promote; bounded provisional validation.
The Bandit’s States [25]	Bandit-based protocol-state selection.	Prior bandit formulation; independent reward and queue evidence here.
Zhang et al. [26] SSGFuzz [27]	Thompson-sampling seed scheduling. State-significance-guided fuzzing.	Reused Beta estimation and Thompson sampling. Code-productivity calibration, distinct from state importance.
Magma [34]; evaluation methodology	Ground-truth bug conditions; repeated evaluation.	Information-controlled network rediscovery; minimal security-patch pairs.

ground-truth bug conditions motivate distinguishing bug-related reachability from triggering [34]; our historical vulnerability protocol additionally requires patch validation and controlled information exposure.

Evaluation methodology is a further boundary. ProFuzzBench and FuzzBench provide structured evaluation settings for protocol and general-purpose fuzzers [35, 36]. Reliability analyses and prudent evaluation practices require care when interpreting coverage, repeated trials, and resource controls [37, 38]. Our protocol adopts these concerns while keeping available retrospective observations distinct from the unfinished confirmatory comparisons. Candidate-level, episode-level, and controlled rediscovery measurements remain necessary to establish the proposed mechanisms.

4. Evidence-Calibrated Controller

This section defines the admission and calibration criteria. Section 5 examines whether the available evidence establishes that these criteria were satisfied. A formal decision rule alone does not demonstrate experimental adherence.

4.1. LLM proposal interface

The model receives permitted RFC fragments, seed traces, and observed interaction summaries and returns candidate requests or bounded scheduling hints. Each candidate has a persistent identifier, source state, intended target/frontier, and links to its prompt and raw reply. Message-field hypotheses remain proposals rather than verified protocol grammars.

The information policy excludes CVE identifiers, security advisories, exploit inputs, and patch contents. It specifies the documentation and observed responses available to the model. Adaptive campaigns may produce different prompts despite identical information policies; prompts and responses must therefore remain traceable.

Outputs that propose requests pass through admission. Outputs that only guide scheduling require structural, state-consistency, and seed-consistency checks and cannot insert durable inputs. Parsing or transport failure does not create a positive admission decision. All attempts, retries, failures, and successful replies enter the declared call/token accounting.

4.2. Admission predicates and evidence separation

For candidate x , $P(x)$ checks the prescribed format, message structure, required fields, lengths, and terminators before execution. A bounded trial then evaluates response acceptability $U(x)$ and reachability $R(x)$: arrival at a declared target/frontier or escape from a recorded rejection sink. Table 3 gives observable evidence and decision criteria.

HTTP/FTP 4xx/5xx classes are rejection priors, not a universal semantic definition. An adapter must specify exceptions before a run and retain the response and reason code. Timeout, reset, disconnect, or an unparseable response gives no positive U evidence. A valid response alone is insufficient for R . Evidence of security violations is retained independently even when normal request admission fails.

Gain is split into

$$G_{\text{code}}(x) = \mathbf{1}\{\text{new code coverage or a favored code-coverage entry}\}, \quad (1)$$

$$G_{\text{state}}(x) = \mathbf{1}\{\text{new IPSM node, state edge, or path}\}. \quad (2)$$

Favored status must be grounded in code coverage; merely being queued or state-novel does not qualify. Source-level code branches are the main evaluation surface. If runtime instrumentation observes code edges instead, that reward definition must be prespecified and reported separately. Hit-count novelty is a separate sensitivity condition. Pre-trial coverage and IPSM snapshots establish which gains are new.

4.3. Provisional and durable queues

Let Q_{prov} contain temporary candidates and Q_{dur} contain long-term mutation seeds. Figure 2 summarizes the three admission outcomes:

$$D(x) = \begin{cases} \text{durable,} & P \wedge U \wedge R \wedge G_{\text{code}}, \\ \text{provisional,} & P \wedge U \wedge R \wedge \neg G_{\text{code}} \wedge G_{\text{state}}, \\ \text{reject,} & \text{otherwise.} \end{cases} \quad (3)$$

IPSM novelty alone is never sufficient for durable promotion. These logical conditions do not assume statistical independence among predicates.

A provisional entry receives at most $H = 64$ descendant executions and a local validation deadline of 30 seconds, whichever

Table 3: Admission predicates and required evidence.

Predicate	Positive evidence	Example failure	Required evidence
P	Prespecified structure and format.	Missing field; invalid length/delimiter.	Candidate identity; validation criteria; reason; hypothesis version.
U	Adapter accepts response.	Rejection; timeout/reset/drop; unparseable reply.	Response identity; acceptance criteria; failures; exceptions.
R	Target/frontier reached or documented sink escape.	Repeated sink; missed destination.	Source/target; observed state sequence.
G_{code}	New code branch/code edge or code-supported favored entry.	IPSM-only novelty.	Pre/post code coverage; novelty class; favored reason.
G_{state}	New IPSM node, state edge, or path.	Known IPSM structure.	Pre/post IPSM; new transitions.

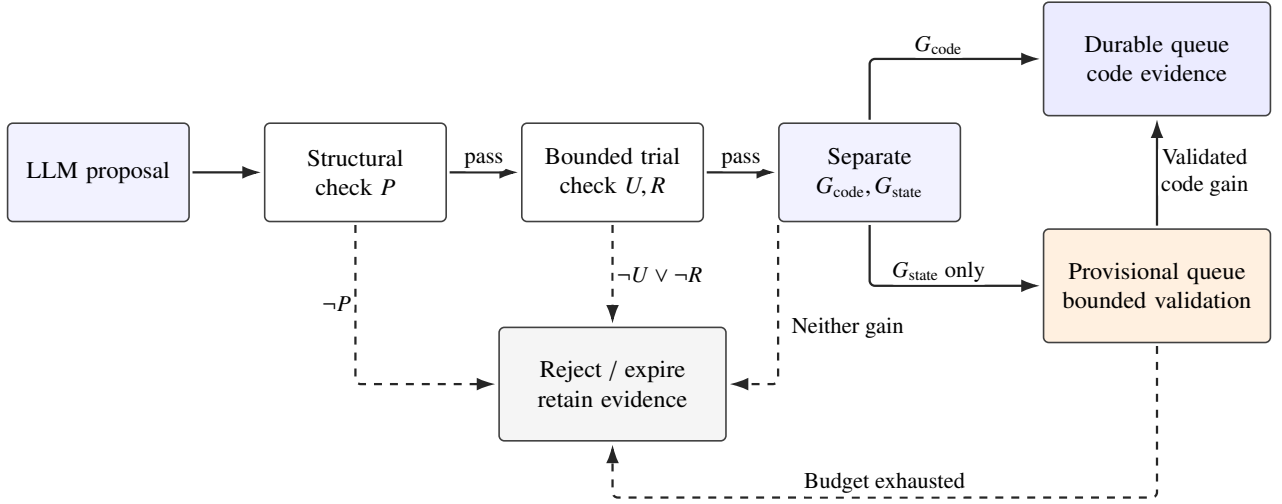


Figure 2: Evidence-gated admission with provisional and durable queues.

is exhausted first; the proposed live provisional capacity is 64. These initial settings are checked in the pilot and prespecified before formal runs. Budget checks occur before each descendant execution. Exhaustion expires the entry from active scheduling while preserving the observations supporting that decision. Campaign termination before the validation horizon is censoring, not unproductive expiration.

A descendant supplying qualifying code evidence permits promotion of the replayable parent sequence and productive descendant, subject to the same structural, response, and reachability checks. The record identifies the first productive descendant and the candidate whose status changes. A global code-discovery event is credited once through execution/lineage IDs, not repeatedly to multiple ancestors.

Productivity estimates select where to propose and continue exploration; they do not independently relax Equation 3. Fixed and calibrated arms use identical admission rules and provisional budgets. Calibration thus affects admission outcomes indirectly through exploration opportunities, preserving the causal scheduling comparison.

4.4. State-selection episodes and independent reward

An episode begins when the scheduler selects response-derived state s and a seed capable of reaching it. It receives one fixed execution-energy budget. Individual executions within the episode do not generate repeated posterior updates. Mutation

counts, reset failures, elapsed time, and budget exhaustion are measured. Infrastructure-interrupted episodes are incomplete, rather than silently assigned reward zero.

For completed episode t targeting s_t , let ΔB_t count previously unseen code branches discovered within that episode. The main reward is

$$r_t = \mathbf{1}\{\Delta B_t > 0\}. \quad (4)$$

A declared code-edge instrumentation variant is admissible, but IPSM novelty is excluded. Favored descendants are a secondary sensitivity reward. Equal energy prevents a state appearing more productive merely because it receives more executions. Trial execution, ordinary mutation, and provisional validation use distinct event types to avoid duplicate updates.

θ_s denotes the probability of positive code reward conditional on selecting s under the episode policy and current campaign phase. It is instrumental utility, not the probability that s is a true semantic state. U continues to mean response acceptability; utility is never denoted $U(s)$.

4.5. Discounted productivity estimates

Each state starts with $\alpha_s = \beta_s = 1$ and

$$\theta_s \sim \text{Beta}(\alpha_s, \beta_s), \quad \widehat{\theta}_{s,t}^{\text{pre}} = \frac{\alpha_{s,t}}{\alpha_{s,t} + \beta_{s,t}}. \quad (5)$$

For a completed selected-state episode:

$$\alpha_s \leftarrow 1 + \gamma(\alpha_s - 1) + r_t, \quad (6)$$

$$\beta_s \leftarrow 1 + \gamma(\beta_s - 1) + (1 - r_t). \quad (7)$$

The proposed discount is $\gamma = 0.995$, with sensitivity values $\{0.99, 0.995, 1.0\}$. Parameters are prespecified after the pilot and before inspecting formal results. At $\gamma = 1$, updates reduce to cumulative Beta–Bernoulli counts. For $\gamma < 1$, excess prior counts are discounted to respond to changing productivity as coverage saturates.

The term productivity posterior denotes this discounted working model. With discounting it is not the exact posterior of a stationary Bernoulli process. Discounting occurs on completed episodes for the selected state; unselected states do not automatically age in wall-clock time. The update clock and cold start are therefore explicit applicability limits.

Every calibration metric uses the pre-update mean recorded before reward. Post-update predictions would leak the outcome into their own evaluation. Thompson samples are recorded separately and are not substituted for predicted Bernoulli probabilities in reliability calculations.

4.6. Calibrated scheduling

The topology term favors underexplored observed states:

$$F(s) = \begin{cases} 4, & s \text{ is newly reachable or } \deg^+(s) \leq 1, \\ 2, & 1 < \deg^+(s) \leq 3, \\ 1, & \deg^+(s) > 3. \end{cases} \quad (8)$$

Only reachable states with eligible seeds participate. $\deg^+(s)$ counts outgoing IPSM state edges, not code branches. Let $B(s)$ denote the inherited base weight shared by C, D, and E. Define

$$\text{Frontier}(s) = B(s)F(s). \quad (9)$$

Keeping this factor and any warm-up policy fixed prevents calibration from simultaneously changing the baseline scheduler. Using the Thompson-sampling principle [39], draw

$$\tilde{\theta}_s \sim \text{Beta}(\alpha_s, \beta_s) \quad (10)$$

and calculate

$$S(s) = \text{Frontier}(s)[\epsilon + (1 - \epsilon)\tilde{\theta}_s], \quad \epsilon = 0.1. \quad (11)$$

Select a state with probability proportional to $S(s)$, recording the pseudorandom seed. This is Thompson-sampled weighted selection, not an argmax policy. Under a finite eligible set, the positive floor gives nonzero selection probability, not a fixed number of selections.

The fixed-policy comparator retains the original penalty:

$$S_D(s) = \text{Frontier}(s)\rho_{\text{fixed}}(s). \quad (12)$$

Its cached queue-yield ratio $p_s = D_s/(N_s + 1)$ counts interesting queued paths D_s per state selections N_s . It is not a calibrated Bernoulli probability and can reflect state-only novelty or unequal execution energy. E replaces only ρ_{fixed} with the calibrated factor in Equation 11, preserving the shared base, admission, prompts, triggers, repair settings, and caps. Integer score conversion and the initial round-robin schedule must be prespecified and verified before comparison.

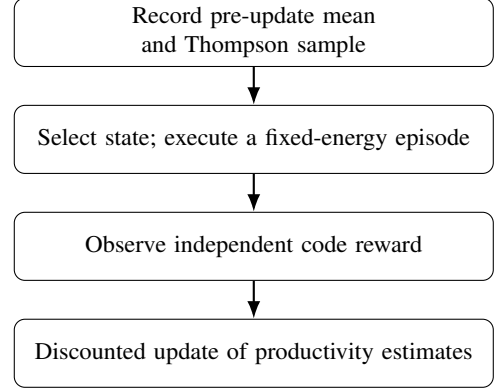


Figure 3: Online productivity calibration using pre-update predictions and code rewards.

Algorithm 1: Evidence-gated candidate processing

Input: Candidate x , target, fixed pre-trial state

- 1 Record candidate, prompt, reply, and lineage identifiers
- 2 **if** $\neg P(x)$ **then**
- 3 Record structural rejection; stop processing x
- 4 Execute one bounded trial without durable queue insertion
- 5 Record $U, R, G_{\text{code}}, G_{\text{state}}$ and reason codes
- 6 **if** $P \wedge U \wedge R \wedge G_{\text{code}}$ **then**
- 7 Promote x to Q_{dur}
- 8 **else if** $P \wedge U \wedge R \wedge \neg G_{\text{code}} \wedge G_{\text{state}}$ **then**
- 9 Insert x into Q_{prov}
- 10 **while** budget, validation deadline, and campaign time remain **do**
- 11 Check budget before the next descendant execution
- 12 Execute and record one descendant
- 13 **if** qualifying code gain and validation pass **then**
- 14 Record first productive descendant; promote; end validation
- 15 Record expiration or campaign censoring if still provisional
- 16 **else**
- 17 Record rejection
- 18 Preserve security-oracle evidence separately

4.7. Intervention policy and optional repair

Plateau interventions follow prespecified detection, timing, and retry rules. The baseline trigger combines IPSM state-edge growth with a sequence of unproductive selections. This dependence on the state proxy is held constant across the causal comparisons; scheduling calibration does not itself validate the intervention trigger.

Primary comparisons of C, D, and E disable repair. Counterexample-guided repair is an optional diagnostic condition. Its evaluation relates the failure and revision to validation outcomes, post-repair predicates, descendant code gain, and token and latency costs. Sparse examples do not establish effectiveness; an independent contribution requires reproducible gains and statistical support across several targets.

Table 4: Operational criteria and evidence limitations.

Requirement	Current boundary	Evaluation consequence
Gate precedes durable promotion	Retention may precede U/R verification.	Disposition labels do not verify queue protection.
Independent state-only trial evidence	State assessment depends on prior retention.	Zero provisional events do not establish a low state-only rate.
Full structural and reachability checks	P : structural recognition; R : generic transition.	Full validity and target reachability remain unverified.
Main reward is new code coverage	Observed novelty may reflect execution frequency.	Separate code branches, execution frequency, and retention.
Fixed-energy episode	Variable mutation energy; zero-execution attempts.	Raw episodes do not provide equal opportunities.
Strict provisional budget	Limits checked after whole fuzzing rounds.	64-mutation/30-second limits may be exceeded.
Verified run configuration	Effective settings may differ from initial declarations.	Verify conditions throughout each experimental trial.
Equal total call/token caps	64-call cap covers plateau use; token enforcement unverified.	Total-budget fairness remains unverified.
Repair disabled in C/D/E	Actual repair conditions are incompletely established.	Reported arm identity does not establish adherence.

5. Operational Validity and Measurement

The evaluated system can represent code and state novelty separately, provisionally retain candidates, and condition state selection on productivity estimates. The available observations do not establish that these mechanisms operated as specified in every trial. Operational validity therefore depends on both effective experimental conditions and the sequence of admission decisions.

Candidate retention may precede behavioral verification. A subsequent U/R failure need not reverse that decision, so strict adherence to Equation 3 is not established. Coverage-based retention may also reflect execution-frequency changes rather than newly covered code branches. Similarly, evaluating provisional budgets only after an entire mutation cycle can exceed the prescribed limits.

IPSM novelty is assessed after changes associated with candidate retention. A candidate without prior code-based retention may therefore lack an independent state-only novelty assessment. This dependency can suppress the provisional condition by construction. An absence of provisional observations cannot distinguish effective filtering from a missing opportunity for provisional validation.

The available P decisions can reflect structural recognition without establishing full request validity. Likewise, R may indicate a state transition without confirming arrival at the intended target or frontier. These observations are weaker than the proposed predicates. Table 4 summarizes the resulting threats to operational validity.

5.1. Measurement structure and traceability

The measurement framework distinguishes experimental conditions, candidate generation, admission trials, provisional outcomes, scheduling episodes, and security outcomes; optional repair adds a seventh observation class. Table 5 specifies the

corresponding evidence. Each observation retains its trial identity, measurement-definition version, temporal order, and time. CVE identities remain isolated benchmark metadata. Missing observations are not assigned zero values.

Confirmatory evaluation requires at least 99.9% completeness of candidate observations, an identifiable candidate for every trial, an identifiable episode for every posterior update, and reconstructable promotion decisions. The denominator includes unsuccessful generation and interpretation attempts. Observation availability, correspondence across measurements, and agreement between reported decisions and actual queue membership are assessed separately.

6. Experimental Protocol

The following protocol specifies the unfinished confirmatory evaluation. The retrospective data do not establish prior registration, prespecified parameters, or completed pilot validation. Remaining experiments require a dated protocol and declared deviations; exploratory and pilot observations are excluded from confirmatory inference.

6.1. Research questions

1. **RQ1: State/code alignment.** How reliably does response-derived state novelty predict downstream code progress across targets and campaign phases? Measure IPSM state-edge/code-branch alignment, episode productivity, calibration, and boundary cases.
2. **RQ2: Admission.** Does execute-before-promote reduce durable queue pollution and improve downstream productivity under matched proposal and scheduling conditions? Compare D with C.
3. **RQ3: Calibration.** Does online productivity calibration improve scheduling over a fixed response-state heuristic? Compare E with D.

Table 5: Observation classes and required measurements.

Observation class	Experimental context	Measurements / outcomes
Experimental conditions	Arm; target/fuzzer versions; execution environment; model/service version; sampling parameters; random seed.	Call/token limits; CPU/memory allocation; observation period; termination reason.
Candidate generation	Candidate and prompt identity; parentage; original request/response; action; source/intended states.	Input/output tokens; generation time.
Admission trial	Candidate/trial correspondence; $P/U/R$ and reasons; response identity; failures; state sequence; target reachability.	Pre/post IPSM nodes/state edges and code coverage; coverage-based priority; latency; decision.
Provisional validation	Candidate identity; validation budget; first productive descendant.	Descendant executions/code gain; conversion time; promotion latency; expiration/censoring.
State-selection episode	Episode/state/seed identity; pre-update $\alpha, \beta, \hat{\theta}$; Thompson sample; frontier/selection scores.	Mutation count; code-branch/IPSM state-edge gain; reward; post-update α, β ; duration; completion.
Security outcome	Anonymous oracle identity; reproducing input; diagnostic evidence; root cause.	Reached/triggered; first trigger time; vulnerable/patched outcomes.
Repair (optional)	Repair lineage; failure/counterexample; original/revised hypothesis; changed component.	Pre/post validation; post-repair predicates/descendant gain; tokens/latency.

Table 6: Required configurations for arms A–E.

Arm	LLM	Admission	Scheduler	Role
A: AFLNet	No	Original AFLNet	AFLNet	Non-LLM baseline.
B: Controlled ChatAFL	Yes	ChatAFL direct/open-loop	ChatAFL lineage	B–A: controlled LLM-lineage change.
C: LoopFuzz-direct	Yes	Parseable candidates directly durable	Fixed LoopFuzz	Admission counterfactual.
D: LoopFuzz-gated-fixed	Yes	Evidence gate + provisional queue	Same fixed LoopFuzz	D–C: admission effect.
E: LoopFuzz-gated-calibrated	Yes	Identical to D	Productivity calibration	E–D: calibration effect.

4. **RQ4: Historical vulnerability rediscovery.** Does the controller improve reproducible rediscovery under independent reached/triggered oracles and patch validation?
5. **RQ5: Cost and applicability.** What runtime, throughput, and LLM costs arise, and under which target characteristics does the method fail to help?

The directional hypotheses are improved downstream productivity for D over C and greater area under the code-branch coverage curve (AUC) for E over D. Regressions, null results, and uncertainty are reported per target rather than hidden in pooled means or replaced by increased IPSM state-edge counts.

6.2. Five arms and causal comparisons

C/D share model configuration, prompt construction, trigger, repair setting, scheduler, and budget policy; the intended intervention is admission. E/D share admission and differ only in state selection. Controlled ChatAFL retains its scheduling lineage; replacing it with the LoopFuzz scheduler would change the baseline. B–A is not an isolated admission comparison.

Proposal control has two scopes. Paired diagnostics evaluate the same candidate under direct and gated admission from an identical initial target and campaign state, with equal descendant budgets. This comparison supports a candidate-level counterfactual. Independent adaptive campaigns diverge, so equal policies do not guarantee identical prompts or model outputs. Replay can assess structural validation and admission decisions under matched initial conditions, but cannot establish equivalence to an independently evolving campaign.

6.3. Targets and experimental controls

The primary design contains nine servers: four FTP servers, Exim (SMTP), Live555 (RTSP), Kamailio (SIP), Forked-daapd (DAAP/DACP over HTTP), and Lighttpd1 (HTTP). Table 7 identifies the target protocols and experimental conditions. Unverified software versions and observation procedures remain unspecified; conditions reported for other studies are not attributed to these trials without supporting evidence.

Each causal comparison requires identical software versions, compilation settings, coverage instrumentation, seeds, replay procedures, and reset policies. Trial order is randomized within machine blocks and balanced across machines and time periods. Random seeds, resource allocations, measured CPU consumption, resource contention, and execution rates characterize each trial. Allocated resources do not establish equal consumption; initialization, model waiting, validation, mutation, and termination all contribute to experimental cost. Resource-efficient benchmarking also makes the cost of obtaining reliable comparisons an explicit concern [40]. The proposed analysis therefore distinguishes allocated CPU-hours, measured consumption, and actual search exposure.

Each arm initially receives 10 independent 24-hour runs per target; C/D/E receive 10 more, reaching 20 each. For nine targets and one allocated CPU per run, the stages require 10,800 and 6,480 CPU-hours, totaling 17,280 before overhead/failures. These are planned resources, not measured usage. A four-target pilot covers Forked-daapd, Lighttpd1, LightFTP, and ProFTPD or Pure-FTPd with five arms, three runs, and four to six hours

Table 7: Targets, protocols, and experimental conditions.

Target	Protocol	Target version	Execution environment	Coverage / reset procedure
LightFTP	FTP			
bftpd	FTP			
ProFTPD	FTP			
Pure-FTPd	FTP			
Exim	SMTP			
Live555	RTSP			
Kamailio	SIP			
Forked-daapd	DAAP/DACP			
Lighttpd1	HTTP			

Table 8: Shared LLM settings required for B–E.

Parameter	Prespecified value	Comparability requirement
Model, version, and inference service		Identical resolved version; name alone insufficient.
Temperature by call class		Match each call class across arms.
Top- p / presence and frequency penalties		Explicit values or documented defaults.
Maximum output tokens per attempt		Shared limit; include truncated/failed responses.
Maximum attempted calls / total tokens		Count all call classes and retries.
Timeout and repeated-request policy		Identical policy; account for waiting and failures.
Model-side seed / support status		Record support; no determinism guarantee.
Prompt corpus / template version		Shared information policy; traceable prompts/responses.
Plateau cadence / threshold / reset		Shared intervention and state-growth criteria.

each. Pilot observations are excluded from formal inference.

An intention-to-run analysis accounts for every planned and initiated trial, including completion, timeout, target or fuzzer failure, infrastructure or measurement failure, exclusion, and repetition, with reasons linking repeated attempts. Low coverage never justifies exclusion. Only prespecified infrastructure failures permit repetition. Interrupted observations are not extended to 24 hours; they contribute failure or censoring outcomes according to the estimand, with sensitivity analysis. Restriction to completed trials does not constitute intention-to-run analysis.

6.4. LLM fairness and configuration

The main design uses equal policy and equal ceilings. All LLM arms must use identical model configurations, intervention policies, and resource ceilings. Actual calls and token usage are reported because adaptive trajectories can trigger different numbers of interventions. Unverified shared settings remain unspecified in Table 8.

An output-token limit per request is distinct from a total campaign limit. A recorded token-limit value of zero does not establish equal-token control without evidence of the effective policy. Monetary costs require verifiable usage and applicable prices; unavailable costs remain unspecified.

Secondary comparisons report code-branch gain per 1,000 total tokens, per LLM call, at a common cumulative token budget, and at a common call count. Security costs use CPU-hours and 100,000-token units. Accounting includes unsuccessful generations, retries, validation, and optional repair. A fixed intervention schedule provides a cost-control sensitivity, not a replacement for the adaptive main policy. Ratios with zero denominators are undefined.

6.5. Mechanism and calibration metrics

For every target/arm, report candidate generation, $P/U/R$ first-failure distribution, direct durable and provisional rates, conversion, expiration, promotion latency, queue size, trial latency, descendant executions, code gain per candidate/token, episode productivity, and throughput. Denominators and incomplete/censored events accompany rates.

Downstream productivity uses a fixed descendant horizon rather than admission gain. For a declared cohort C of N candidates enrolled before their outcomes are known, let $Y_i(H) = 1$ if a later descendant produces new code coverage within H executions, and 0 if the full horizon is observed without gain. With complete observation, define

$$\text{Productivity@ } H = \frac{1}{N} \sum_{i \in C} Y_i(H). \quad (13)$$

Initially $H = 64$. The admission trial is excluded. A positive outcome can be established early; a negative requires the complete horizon. With n_+ known positives and $n_?$ censored unresolved candidates, report the cohort bounds $[n_+/N, (n_+ + n_?)/N]$ rather than deleting incomplete negatives from the denominator. The primary paired diagnostic reserves the same full validation horizon for both controllers; incomplete online cohorts are reported with these bounds and censoring counts. Durable and provisional cohorts are distinct; post-promotion productivity uses a fresh horizon. Thus neither admission gain nor selective observation mechanically raises measured productivity.

Avoided queue pollution requires C–D evidence. In the paired diagnostic, identify direct-arm durable candidates without downstream gain and compare their retention under gating. D-arm rejections are not automatically pollution. A random,

predeclared sample of rejected candidates receives shadow validation to estimate false negatives, with inclusion probabilities, snapshots, lineage, and cost recorded.

For complete episodes, $p_t = \hat{\theta}_{s_t,t}^{\text{pre}}$ predicts later reward r_t . The Brier score (BS) [41] is

$$\text{BS} = \frac{1}{n} \sum_{t=1}^n (p_t - r_t)^2. \quad (14)$$

Expected calibration error (ECE) [42] uses ten fixed equal-width probability bins and reports counts, mean predictions, and observed frequencies. Empty bins stay empty. Reliability plots and the area under the precision–recall curve (AUPRC) [43] accompany the BS; with no positive episodes AUPRC is undefined. The empirical diagnostics summarize AUPRC using non-interpolated average precision (AP). Compare the BS to a prequential base-rate predictor because rare rewards can make uninformative near-zero predictions look accurate. IPSM state-edge/code-gain ratios with zero code gain are undefined or shown as separate zero-gain cases.

Episodes within a run are dependent; uncertainty is clustered by run and reported by target/phase. Posterior trajectories and state weights are diagnostic, not evidence that every state is accurately learned. Correlation between IPSM growth and later code gain is evaluated within target and phase, without pooling raw counts across unlike programs.

6.6. Coverage and statistical analysis

Primary coverage outcomes are code-branch AUC over 24 hours and final code-branch count. AUC uses each run’s cumulative time series with a common interpolation convention and instrumentation, in branch-hours. Endpoints recorded at unequal durations are not substituted. Time to 95% of a target-specific common reference endpoint uses a threshold prespecified before formal comparisons; unreached thresholds are censored.

Per target, report mean, median, dispersion, bootstrap 95% confidence intervals (CIs) [44], two-sided Mann–Whitney U tests [45], and Cliff’s δ [46]. The primary family comprises 36 coverage tests across nine targets: D–C and E–D contrasts for final branches and AUC. The Benjamini–Hochberg procedure [47] controls this family at $q = 0.05$. The primary admission-mechanism family has nine D–C productivity contrasts and is reported separately. Other diagnostics and sensitivity tests are exploratory. Effect direction is treatment minus comparator.

Cross-target summaries use equal-weight target effects and a hierarchical bootstrap resampling targets, then runs within targets. Raw branch counts from different programs or individual executions are not pooled as independent samples. Target regressions and uncertainty accompany aggregates. Historical time-to-trigger uses Kaplan–Meier estimates [48], right censoring, and CVE-specific deadline recall. Fuzzer-failure interruptions are distinguished because censoring may be informative.

7. Available Evidence and Results

The evaluation analyzes recorded outcomes from the benchmark and ablation studies. Results obtained under earlier con-

figurations do not establish the effectiveness of the proposed controller. Blank numeric cells indicate unavailable or unvalidated evidence rather than measured zeros.

7.1. Sample composition and observation periods

The dataset contains 94 AFLNet (A), 91 ChatAFL (B), 104 benchmark LoopFuzz (D), 90 direct-admission (C), and 86 calibrated (E) observations. The sensitivity conditions contain 108 observations at $\gamma = 0.99$ and 55 at $\gamma = 1.0$. All available observations are retained without assuming independence. The nominal target is ten trials per condition; actual sample sizes n remain explicit. Means and sample SDs use observed values only, with no imputation for unavailable trials.

All 104 benchmark LoopFuzz D observations are reported as completed; Table 9 gives target-specific counts. Termination categories alone do not establish memory exhaustion or vulnerability activation. Reported runtime measures active fuzzing rather than total CPU consumption or the entire experimental period.

The benchmark LoopFuzz observations also constitute the fixed-policy admission condition D and are counted once. The direct-admission and calibrated-scheduling conditions correspond to C and E. Two sensitivity groups use $\gamma = 0.99$ and $\gamma = 1.0$. The configuration uncertainty in Table 4 and unequal sample sizes preclude a fully matched comparison. Recorded posterior estimates alone do not establish a calibration effect.

The complete target–arm results, including both gamma sensitivity groups, are reported in Appendix A (Table A.1).

Table 10 includes every available A and D observation. Code-branch and IPSM state-edge endpoints represent the terminal measurements from each trial. For all 628 experimental records, the final trajectory measurement agrees with the corresponding endpoint. These are descriptive observations at unequal durations, rather than matched 24-hour outcomes. Sample SDs describe variation among the available observations; no causal effect or significance claim is attached to the endpoint differences.

7.2. RQ1: Current alignment observations

Forked-daapd has mean code-branch endpoints of 2330.8 ± 60.0 for AFLNet and 2670.2 ± 87.3 for LoopFuzz, while IPSM state-edge endpoints are 21.6 ± 2.1 and 22.8 ± 2.9 . This direction does not reproduce the older motivating claim of code regression. The difference cannot be attributed to calibration, which is not established as active.

Lighttpd1 increases in both endpoint metrics: code branches from 1940.7 ± 21.2 to 2126.7 ± 15.7 , and IPSM state edges from 17.7 ± 2.5 to 28.9 ± 2.7 . These aggregate observations do not demonstrate the hypothesized small state expansion with hidden code progress. LightFTP has near-equal code endpoints, 71.0 ± 0.0 versus 71.7 ± 2.3 , with larger LoopFuzz IPSM state-edge counts, 187.3 ± 11.3 versus 207.0 ± 17.0 . That pattern motivates a saturation analysis but does not establish the full time course or identify individual misleading states.

All three cases require common experimental versions, matched observation periods, and linked state/execution measurements before supporting a mechanism claim. Figure 4

Table 9: Available observations by target; D-arm status and duration.

Target	AFLNet	ChatAFL	LoopFuzz(D)	Completed(D)	Exit 137(D)	Duration (min)
LightFTP	13	11	15	15	0	1473–1522
bftpd	10	10	10	10	0	1499–1510
ProFTPD	10	10	12	12	0	1489–1518
Pure-FTPd	11	10	10	10	0	1460–1472
Exim	10	10	10	10	0	1477–1486
Live555	10	10	12	12	0	1489–1505
Kamailio	10	10	12	12	0	1472–1518
Forked-daapd	10	10	12	12	0	1474–1524
Lighttpd1	10	10	11	11	0	1469–1523

Table 10: A/D code and IPSM endpoints (mean $\pm$ SD).

Target	Code branches		IPSM state edges	
	AFLNet	LoopFuzz(D)	AFLNet	LoopFuzz(D)
LightFTP	71.0 $\pm$ 0.0	71.7 $\pm$ 2.3	187.3 $\pm$ 11.3	207.0 $\pm$ 17.0
bftpd	486.0 $\pm$ 8.8	502.7 $\pm$ 7.5	163.1 $\pm$ 15.4	226.7 $\pm$ 14.0
ProFTPD	5053.8 $\pm$ 97.4	5428.1 $\pm$ 104.0	190.5 $\pm$ 27.9	305.3 $\pm$ 20.3
Pure-FTPd	1076.4 $\pm$ 89.6	1223.3 $\pm$ 29.7	236.5 $\pm$ 26.7	321.4 $\pm$ 11.1
Exim	3798.8 $\pm$ 43.3	3914.9 $\pm$ 20.9	88.8 $\pm$ 5.2	152.7 $\pm$ 9.3
Live555	2893.6 $\pm$ 14.5	3067.1 $\pm$ 29.5	96.1 $\pm$ 8.6	170.1 $\pm$ 11.1
Kamailio	10002.6 $\pm$ 144.2	10631.2 $\pm$ 472.9	109.4 $\pm$ 9.7	128.2 $\pm$ 26.6
Forked-daapd	2330.8 $\pm$ 60.0	2670.2 $\pm$ 87.3	21.6 $\pm$ 2.1	22.8 $\pm$ 2.9
Lighttpd1	1940.7 $\pm$ 21.2	2126.7 $\pm$ 15.7	17.7 $\pm$ 2.5	28.9 $\pm$ 2.7

Table 11: D-arm observations; reported decisions do not verify queue membership.

Target	Candidates	Trials	Reject	Provisional	Durable	Episodes
LightFTP	48	48	44	0	4	13604
bftpd	13	13	13	0	0	3249
ProFTPD	0	0	0	0	0	2395
Pure-FTPd	173	173	166	0	7	5378
Exim	24	24	24	0	0	2650
Live555	851	851	844	0	7	7687
Kamailio	253	253	249	0	4	9004
Forked-daapd	412	411	410	0	1	12056
Lighttpd1	453	453	453	0	0	21570

instead examines a narrower estimand: agreement between pre-update probabilities and the binary reward associated with coverage-based retention, separately by target. All 31,826 E observations satisfy probability, Beta-mean, binary-response, uniqueness, temporal-order, and within-episode discounted-update consistency checks. Excluding 6,070 episodes with no mutations leaves 25,756 episodes from 86 runs. Three discontinuities between successive posterior estimates are retained; missing history is not reconstructed. These measurements alone do not verify that every run followed the intended scheduling policy.

Table 12 reports the BS, ten-bin ECE, and AUPRC as AP; lower BS/ECE is better. Its prequential reference uses only past included same-run rewards with Beta(1,1) smoothing. Percentile 95% CIs use 2,000 bootstrap resamples of whole runs within each target; they do not resolve unknown dependence between launches. Empty bins have no estimate, and bins represented by fewer than two runs have no CI. In Figure 4, points pair the bin-

mean pre-update prediction with reward frequency; the diagonal denotes perfect calibration. Lower strips give positive-mutation episode counts, with actual runs n , episodes m , and nominal $N = 10$. The observed reward reflects coverage-based retention, which may include execution-frequency novelty; positive mutation counts do not establish complete fixed-energy episodes. Consequently, these diagnostics do not fill the independent code-branch calibration outcomes in Table 14.

7.3. RQ2: Observed admission records and missing counterfactual

The D condition contains 2,227 candidate observations and 2,226 validation trials, with 2,203 reported rejections, 23 durable admissions, and no provisional admissions. One candidate has no corresponding trial; every trial has a corresponding candidate. C and E contain 1,326/1,326 and 735/734 candidates/trials, respectively. C has 1,326 reported durable admissions; E has 725 rejections and 9 durable admissions. These reported decisions

Table 12: E-arm proxy-reward calibration (95% CIs in brackets).

Target	n	Episodes	BS	Reference BS	ECE (10 bins)	AUPRC (AP)
LightFTP	10	6,622	0.188 [0.178, 0.199]	0.183 [0.173, 0.196]	0.166 [0.161, 0.172]	0.452 [0.432, 0.472]
bftpd	10	2,643	0.249 [0.243, 0.257]	0.247 [0.243, 0.251]	0.105 [0.094, 0.118]	0.517 [0.493, 0.541]
ProFTPd	10	1,002	0.183 [0.179, 0.188]	0.144 [0.126, 0.157]	0.137 [0.106, 0.173]	0.810 [0.782, 0.846]
Pure-FTPd	10	2,848	0.227 [0.218, 0.234]	0.243 [0.233, 0.248]	0.053 [0.041, 0.071]	0.700 [0.674, 0.721]
Exim	10	1,057	0.230 [0.222, 0.239]	0.219 [0.211, 0.227]	0.126 [0.107, 0.149]	0.703 [0.674, 0.733]
Live555	10	2,259	0.224 [0.212, 0.233]	0.218 [0.206, 0.229]	0.082 [0.065, 0.104]	0.742 [0.713, 0.771]
Kamailio	10	6,374	0.163 [0.153, 0.171]	0.161 [0.149, 0.173]	0.042 [0.032, 0.056]	0.858 [0.847, 0.866]
Forked-daapd	6	754	0.246 [0.236, 0.258]	0.252 [0.249, 0.256]	0.093 [0.063, 0.133]	0.622 [0.587, 0.643]
Lighttpd1	10	2,197	0.231 [0.225, 0.237]	0.251 [0.249, 0.253]	0.085 [0.072, 0.097]	0.665 [0.638, 0.687]

Pre-update probability and observed proxy reward

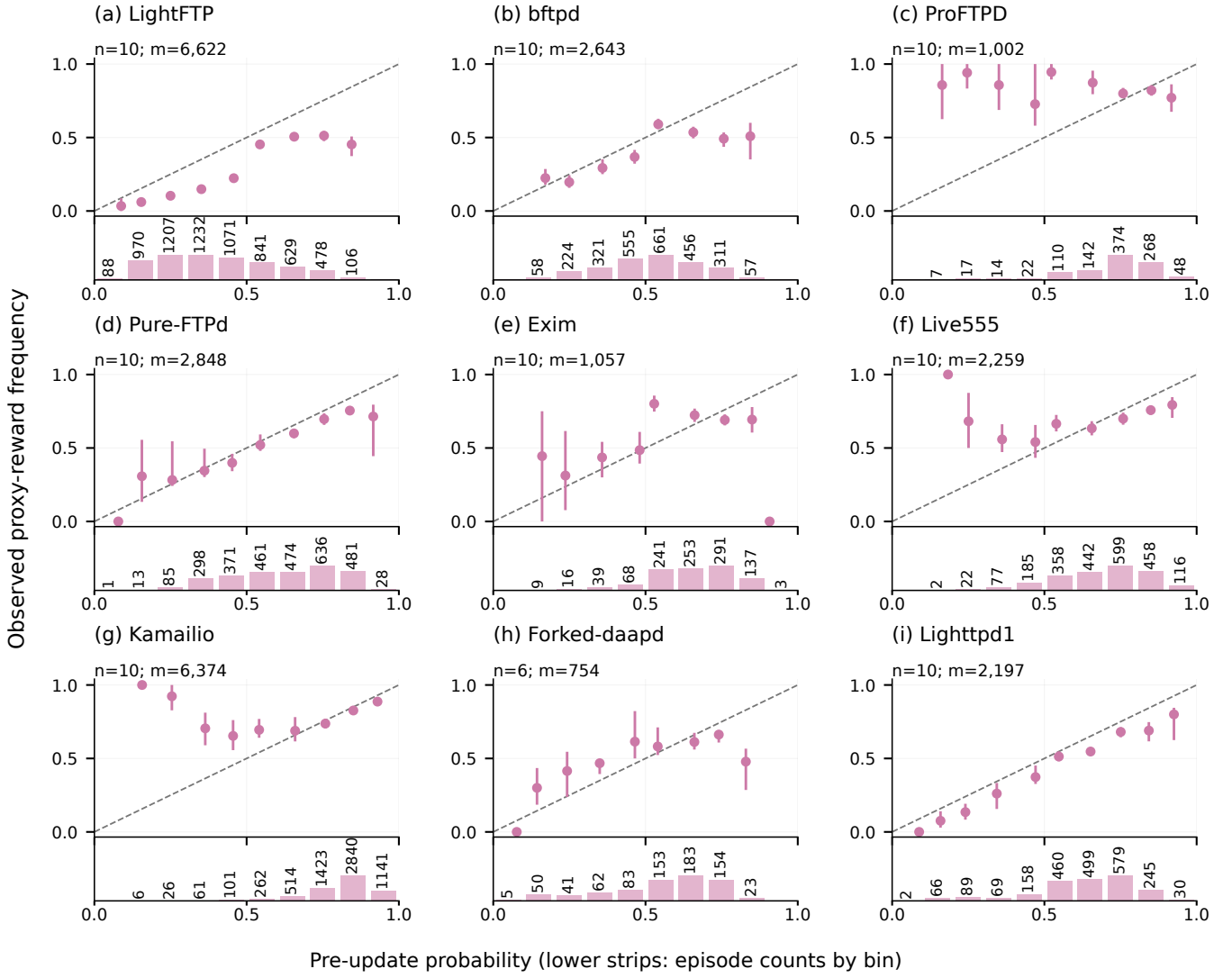


Figure 4: E-arm proxy-reward calibration. Points: bin means; bars: 95% CIs.

do not independently verify queue membership or establish the required 99.9% observation completeness.

Candidate and admission observations are available for C, D, and E, with target-specific gaps. ProFTPD lacks these observations in D and E despite nonzero model usage. Provisional-outcome and repair-outcome measurements are unavailable. Table 11 gives D-condition counts, and Table A.1 gives endpoint results. Absence of observations does not establish zero conversion rates or absence of provisional validation and repair.

Reported dispositions do not independently establish durable queue membership. Retention before verification and incomplete state-only novelty assessment prevent interpreting rejections as avoided queue pollution or durable labels as predicate conformance. Without a validated C–D comparison and descendant lineage, downstream productivity, false-negative rates, and avoided nonproductive retention remain unmeasured. Table 14 and Figure 7 summarize the available decisions and latency measurements descriptively. A first-failure indicator does not by itself explain actual retention or rejection.

7.4. RQ3: Scheduling evidence and unfilled causal results

C, D, and E contain 56,164, 77,593, and 31,826 state-selection episodes, respectively. Mutation counts range from zero to 18,816 per episode, precluding an equal-energy interpretation. Some coverage observations represent distinct instrumentation locations or coverage-based retention events rather than independent source-level code branches. They therefore cannot be combined with terminal code-branch counts or interpreted as the primary reward in Equation 4.

The evaluation includes the calibrated condition E and two discount-factor sensitivity conditions. Heterogeneous observation periods and uncertain effective scheduling policies prevent a matched causal interpretation of E–D. Table 13 reports the recoverable endpoints and AUCs; all 45 main target–arm combinations are represented, including ten E/bftpd observations. Neither sensitivity condition contains Forked-daapd observations. A calibration benefit requires a prespecified analysis under matched observation periods.

Fig. 5 shows code-branch medians and interquartile ranges on a five-minute grid over 0–24 hours from fuzzer startup; the bands are not CIs. Linear interpolation stays within recorded observations, and labels give the contributing n at 24 hours. Table 13 instead integrates each full observed trajectory in branch-hours; cells show branches with actual n above AUC, both as mean $\pm$ SD. Each run contributes only between its first and last observation; time-dependent denominators accompany the figure data. Changes in the set of contributing runs can change an aggregate curve even when individual histories do not decrease. In 127 of the 465 observed trajectories, at least one recorded branch count decreases. The figure preserves these observations without applying a cumulative maximum or otherwise correcting the replay output; both within-run decreases and changing support can produce aggregate decreases.

Fig. 6 presents three reproducibly selected state histories. For each motivating target, the E run is selected by the lower median terminal branch count, with a deterministic record ordering resolving ties; its two states with the most positive-mutation

episodes are shown. Selection uses neither calibration accuracy nor effect direction. Complete histories, including zero-mutation episodes, show pre-update probabilities above the final selection/frontier-score ratio. Time denotes episode end; lines guide the eye, and colors identify states within each target. Undefined zero-frontier ratios are omitted. These cases illustrate observed behavior without establishing a causal state/code mismatch.

7.5. RQ5: Recorded costs and interpretation limits

Table 15 reports model calls and token usage for D by target; supporting results also give C/E ranges and means. Model calls range from 52 to 248 per D run, so the stated limit of 64 does not establish a campaign-wide ceiling. Usage estimates have not been independently reconciled with provider charges and may omit the final observation interval.

The reported conditions identify a model name, grammar/plateau temperatures of 0.5/1.2, top- p of 1.0, an output limit of 4096 tokens, and a token-limit setting of 0. A model name does not uniquely identify the inference version. Uncertainty in effective conditions and resource measurement limits interpretation; these values do not establish the shared confirmatory settings in Table 8.

Candidate token fields do not necessarily cover all call classes or retries; summing them is not total campaign cost. Low-activation targets, high-rejection targets, token outliers, saturation, and regressions must remain in the final analysis. Low activation may reflect productive mutation or a restrictive trigger; high rejection may reflect either useful gating or false negatives. Both need mechanism evidence.

Figures 8 and 9 plot reported token usage and model calls against terminal code branches for 371 B–E runs, using target-specific scales. Tokens combine prompt and completion counts and are displayed in thousands (k). Comparable AFLNet usage measurements are unavailable; calls measure neither token expenditure nor CPU time. These views reveal dispersion and usage ranges without estimating a causal return per token or call. Measured CPU-hours and controlled A–E CVE recall remain unavailable. The case-level replays in Section 8 do not supply those cost–performance axes.

8. Vulnerability Replay Evidence and Rediscovery Protocol

The vulnerability study comprises four cases: three associated with CVE identifiers and one SMARTPL availability failure without a CVE. Each case has a report and a triggering input; three have direct execution evidence, and ProFTPD also has a reported patched-version comparison. Tables 17 and 18 summarize these observations. Case identities, versions, and replay counts follow the original reports; incomplete patch and experimental-configuration evidence prevents independent verification of every version pair.

8.1. Case identity and evidence provenance

We distinguish a direct execution observation (L) from a replay result stated only in a case report (R). A report can contain additional trials not represented by the available execution

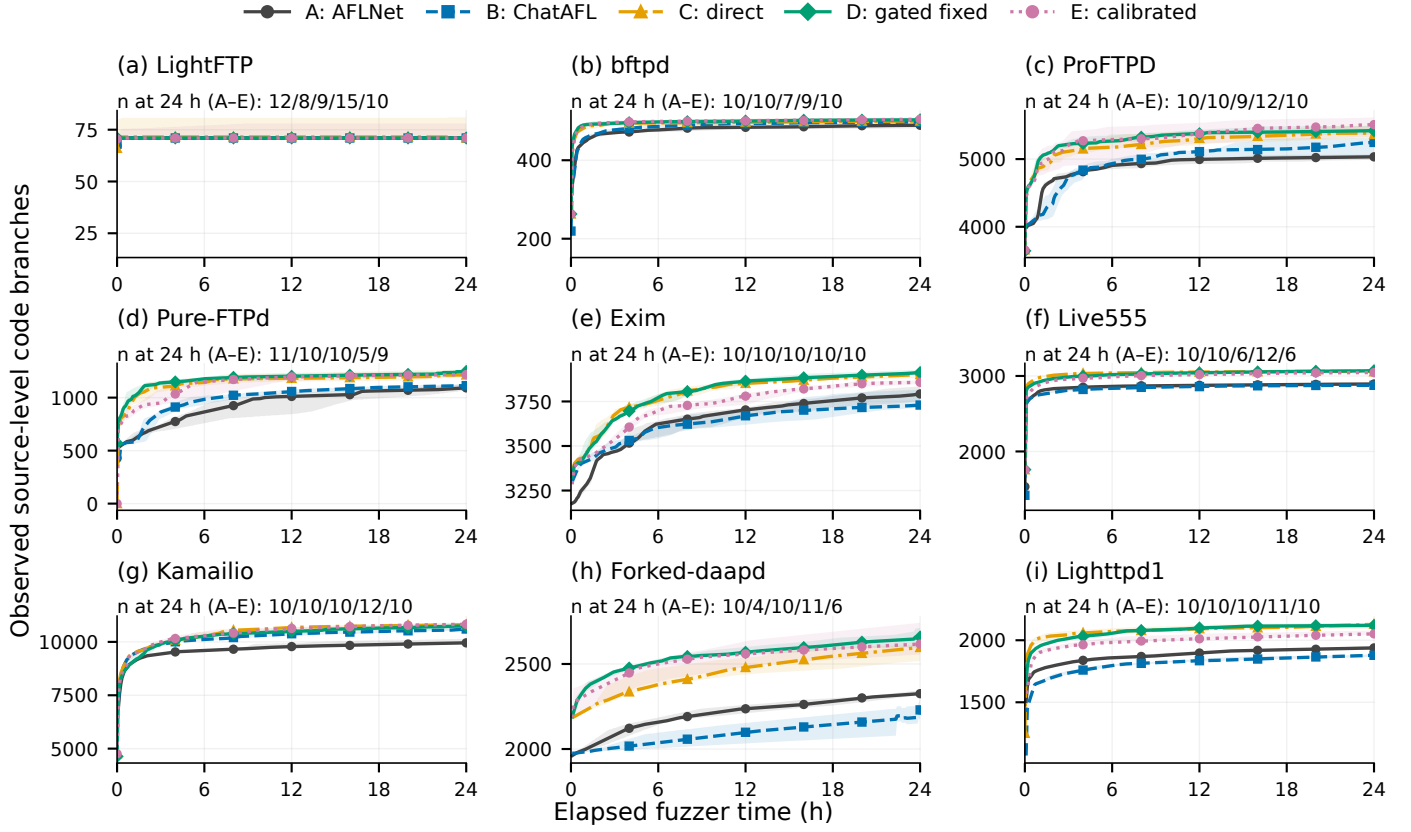


Figure 5: A-E code-branch trajectories: medians and interquartile ranges.

Productivity estimates and scheduling multipliers: three E cases
 Selection: lower-median terminal coverage run, then two most observed states

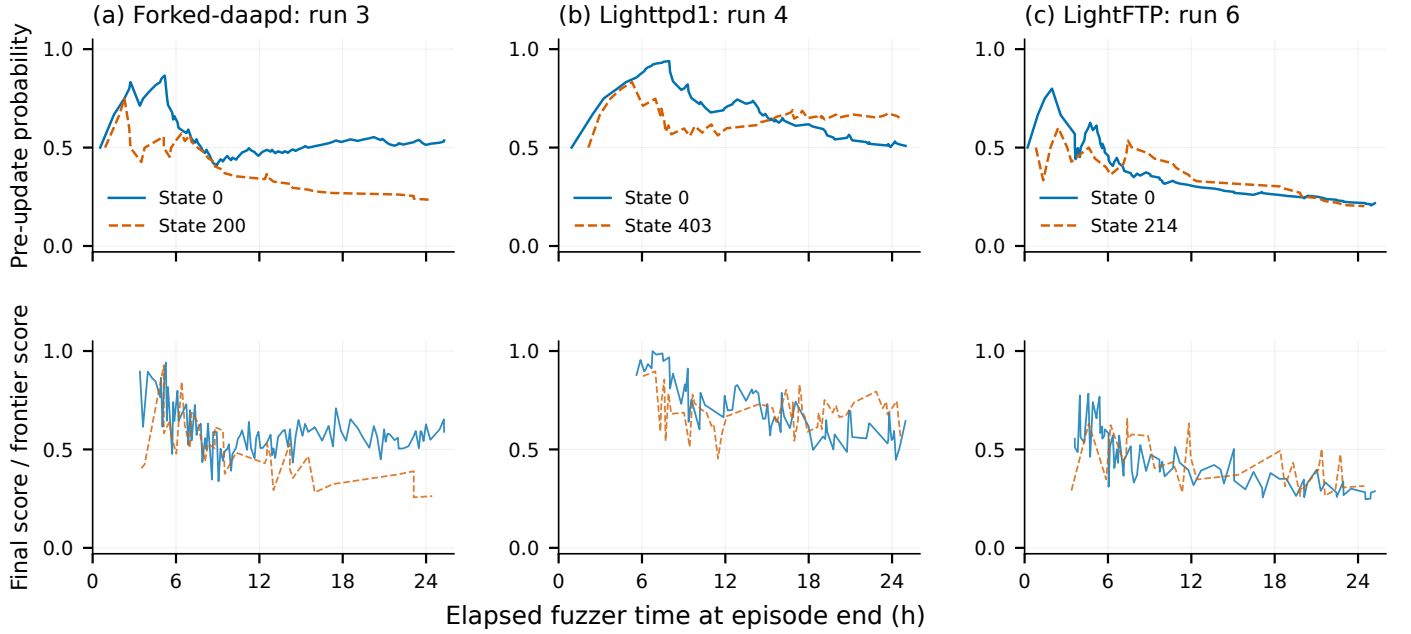


Figure 6: E-arm state histories: pre-update probabilities (top) and scheduling multipliers (bottom).

Table 13: Branch endpoints (top) and AUCs (bottom), mean $\pm$ SD.

Target	A	B	C	D	E
LightFTP	71.0 $\pm$ 0.0 (n=13)	71.3 $\pm$ 0.9 (n=11)	80.6 $\pm$ 15.5 (n=10)	71.7 $\pm$ 2.3 (n=15)	73.9 $\pm$ 4.5 (n=10)
	1758.5 $\pm$ 33.8	1728.1 $\pm$ 36.5	1987.7 $\pm$ 394.1	1780.7 $\pm$ 45.2	1825.0 $\pm$ 107.6
bftpd	486.0 $\pm$ 8.8 (n=10)	492.2 $\pm$ 14.1 (n=10)	498.9 $\pm$ 12.2 (n=10)	502.7 $\pm$ 7.5 (n=10)	503.7 $\pm$ 7.1 (n=10)
	11868.7 $\pm$ 165.6	12068.6 $\pm$ 323.8	12082.0 $\pm$ 451.0	12267.7 $\pm$ 387.9	12416.6 $\pm$ 220.9
ProFTPD	5053.8 $\pm$ 97.4 (n=10)	5205.9 $\pm$ 166.7 (n=10)	5446.3 $\pm$ 194.7 (n=10)	5428.1 $\pm$ 104.0 (n=12)	5469.8 $\pm$ 215.7 (n=10)
	122652.5 $\pm$ 1730.1	125217.7 $\pm$ 2989.1	130538.5 $\pm$ 4511.6	132539.2 $\pm$ 2868.9	130564.7 $\pm$ 4604.1
Pure-FTPd	1076.4 $\pm$ 89.6 (n=11)	1109.3 $\pm$ 40.0 (n=10)	1221.2 $\pm$ 48.9 (n=10)	1223.3 $\pm$ 29.7 (n=10)	1215.4 $\pm$ 63.7 (n=10)
	23054.1 $\pm$ 2084.4	24976.6 $\pm$ 1126.8	28860.8 $\pm$ 1645.7	27863.9 $\pm$ 1496.9	27894.3 $\pm$ 1737.6
Exim	3798.8 $\pm$ 43.3 (n=10)	3761.7 $\pm$ 60.4 (n=10)	3896.1 $\pm$ 25.6 (n=10)	3914.9 $\pm$ 20.9 (n=10)	3858.3 $\pm$ 32.8 (n=10)
	91083.0 $\pm$ 1031.5	91301.7 $\pm$ 1501.4	94880.1 $\pm$ 928.3	93659.0 $\pm$ 573.1	93123.6 $\pm$ 924.6
Live555	2893.6 $\pm$ 14.5 (n=10)	2882.4 $\pm$ 22.0 (n=10)	3075.7 $\pm$ 34.4 (n=10)	3067.1 $\pm$ 29.5 (n=12)	3054.2 $\pm$ 20.5 (n=10)
	71127.4 $\pm$ 517.3	71224.8 $\pm$ 422.5	73511.0 $\pm$ 562.9	75436.1 $\pm$ 712.2	72378.5 $\pm$ 1049.8
Kamailio	10002.6 $\pm$ 144.2 (n=10)	10644.3 $\pm$ 96.5 (n=10)	10800.6 $\pm$ 129.0 (n=10)	10631.2 $\pm$ 472.9 (n=12)	10830.0 $\pm$ 91.3 (n=10)
	242853.7 $\pm$ 2772.6	257606.8 $\pm$ 2082.0	260260.9 $\pm$ 2942.4	254332.1 $\pm$ 9492.1	259377.2 $\pm$ 2649.6
Forked-daapd	2330.8 $\pm$ 60.0 (n=10)	2169.6 $\pm$ 113.0 (n=10)	2585.4 $\pm$ 86.2 (n=10)	2670.2 $\pm$ 87.3 (n=12)	2648.2 $\pm$ 116.0 (n=6)
	54453.2 $\pm$ 1494.1	49393.2 $\pm$ 3335.9	61776.8 $\pm$ 1863.7	63646.7 $\pm$ 1977.0	62349.5 $\pm$ 1726.9
Lighttpd1	1940.7 $\pm$ 21.2 (n=10)	1884.1 $\pm$ 25.3 (n=10)	2145.0 $\pm$ 69.8 (n=10)	2126.7 $\pm$ 15.7 (n=11)	2068.1 $\pm$ 44.5 (n=10)
	46909.4 $\pm$ 235.7	45075.8 $\pm$ 729.3	53003.8 $\pm$ 1613.7	51223.2 $\pm$ 621.1	50385.1 $\pm$ 941.3

Table 14: C/D/E mechanism observations; blanks indicate unvalidated evidence.

Metric	C	D	E	Denominator / interpretation
Candidate records	1,326	2,227	735	Distinct candidates within each experimental record.
Trial records	1,326	2,226	734	Distinct trials within each experimental record.
Reject / durable labels	0 / 1,326	2,203 / 23	725 / 9	Reported trial outcomes.
First failed <i>P/U/R</i>	0 / 1,208 / 0	0 / 2,071 / 0	0 / 687 / 0	First false <i>P</i> , then <i>U</i> , then <i>R</i> ; denominator: trials.
Direct durable rate	100.0% (1,326/1,326)	1.0% (23/2,227)	1.2% (9/735)	Durable labels / generated candidates.
State-only provisional rate				All candidates; disabled for C.
Provisional conversion / expiration				Provisional admissions; include pending/censored cases.
Trial latency (ms)	439.2 $\pm$ 405.9 (<i>n</i> = 67)	403.6 $\pm$ 574.6 (<i>n</i> = 81)	787.3 $\pm$ 552.9 (<i>n</i> = 60)	Mean $\pm$ SD of within-run trial means; contributing runs only.
Promotion latency				Verified queue-promotion timestamps required.
Downstream productivity@64				Descendant code gain; exclude admission gain.
Queue size / nonproductive retention				Queue state and C–D counterfactual.
Shadow-validation false-negative rate				Random rejected-candidate sample.
Code-branch gain per episode				Completed equal-energy episodes.
BS / ECE / AUPRC				Fixed-energy code-branch reward; proxy diagnostics: Table 12.
Code gain per candidate / 1,000 tokens				Joined code gain and complete usage.

Table 15: Reported D-arm costs: per-run ranges by target.

Target	Model calls	Total tokens (k)
LightFTP	59–161	106.0–224.7
bftpd	199–248	198.0–301.8
ProFTPD	188–192	170.1–198.8
Pure-FTPd	207–242	200.2–264.1
Exim	106–123	109.4–151.0
Live555	123–189	264.1–433.1
Kamailio	56–108	157.0–285.5
Forked-daapd	52–163	103.4–540.9
Lighttpd1	131–187	168.5–252.5

Table 16: Optional repair diagnostics; unavailable values remain blank.

Metric	E, repair off	E, repair on
Repair opportunities / attempts		
Before/after validation success		
Post-repair descendant code gain		
Changed fields / locality		
Token / latency cost		
Independent effect / interval		

evidence. Duplicate diagnostic observations do not constitute independent trials. Input identity and byte length are retained in the supporting evidence. This retrospective analysis involves no additional target executions. Table 17 identifies the reported

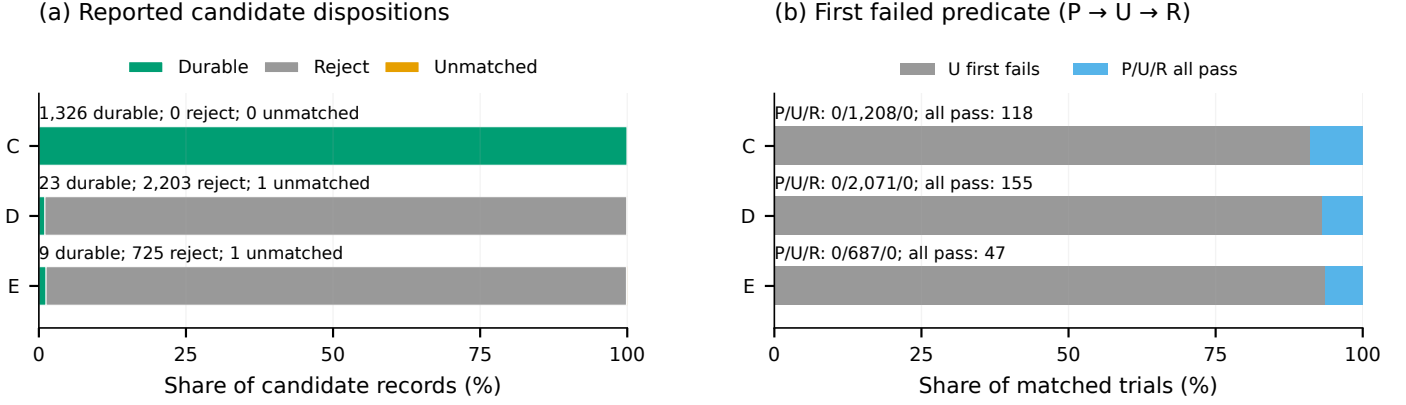


Figure 7: C/D/E reported dispositions (left) and first failed $P/U/R$ predicates (right).

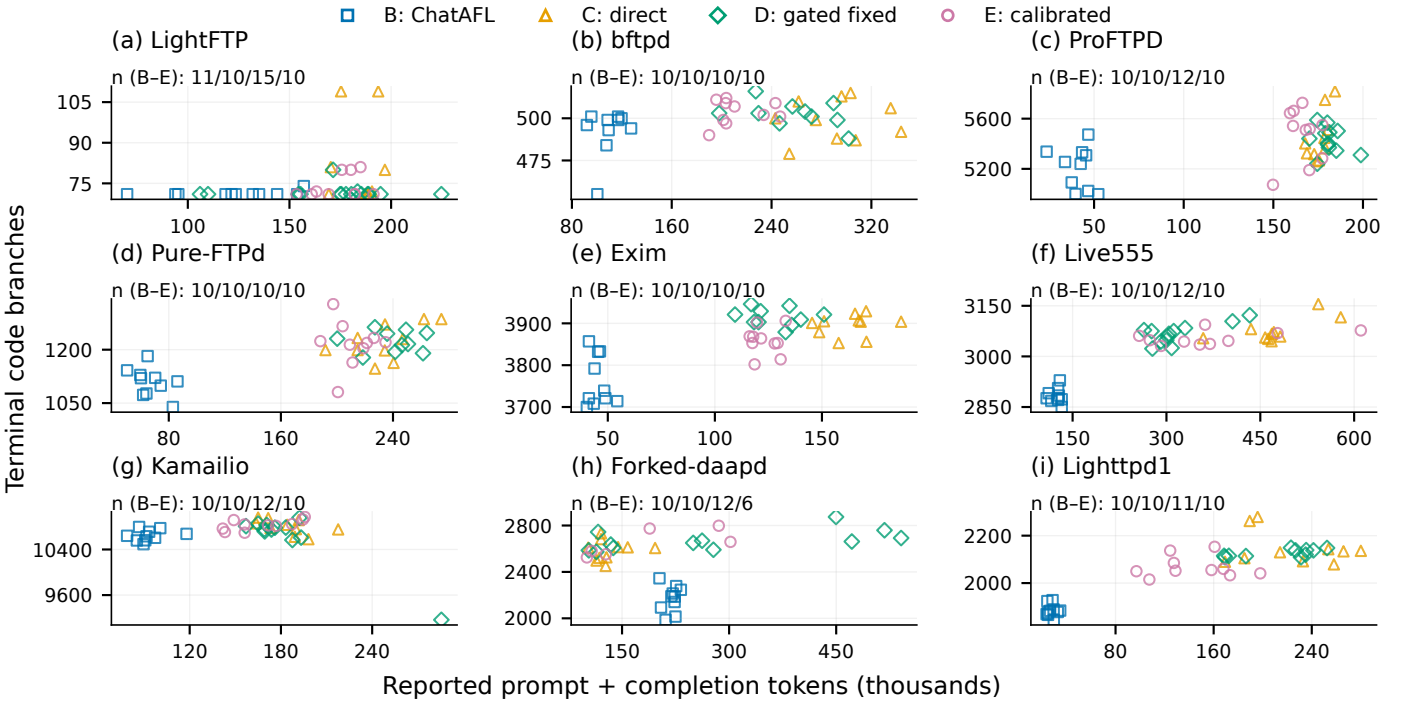


Figure 8: Reported tokens versus terminal code branches, by target and arm.

versions, triggering conditions, and comparative evidence.

For ProFTPd, the reported defect is an out-of-bounds read during FTP command processing, supported by an ASan heap-buffer-overflow observation. The oracle uses a 65,535-byte command buffer; the report states that the default 512-byte configuration does not expose the same error to ASan. The result is therefore conditional on this setting. Three reported patched-version trials return an FTP 500 response without ASan activation and terminate normally. This supports differential replay evidence, although experimental-version identity and exact patch equivalence remain unverified.

For LIVE555, direct diagnostic evidence supports a use-after-free triggered by the public PoC. A duplicated observation is treated as one diagnostic signature rather than independent replication. The report attributes fuzzer-generated inputs to the same memory-lifetime failure, records 8/9 replay hits, and

describes an unsuccessful local patch. The fuzzer-generated reproducer, complete memory-lifetime evidence, and individual trial outcomes are unavailable. The public PoC supports replay validation, but does not demonstrate independent rediscovery by the evaluated controller or independently establish cross-version CVE attribution.

For Kamailio, a negative parsed Content-Length supports the reported integer-overflow condition. The trigger requires TCP and is not assessed by the UDP-only benchmark condition. A local overflow guard reportedly suppresses the diagnostic, but direct patched-version evidence and repetition counts are unavailable. The observation remains report-only and does not establish equivalence to the upstream patch. An earlier unsupported 20/20 versus 0/20 claim is excluded from quantitative results. The SMARTPL case reports reproducible unavailability from a known input, while none of five sampled campaign hangs

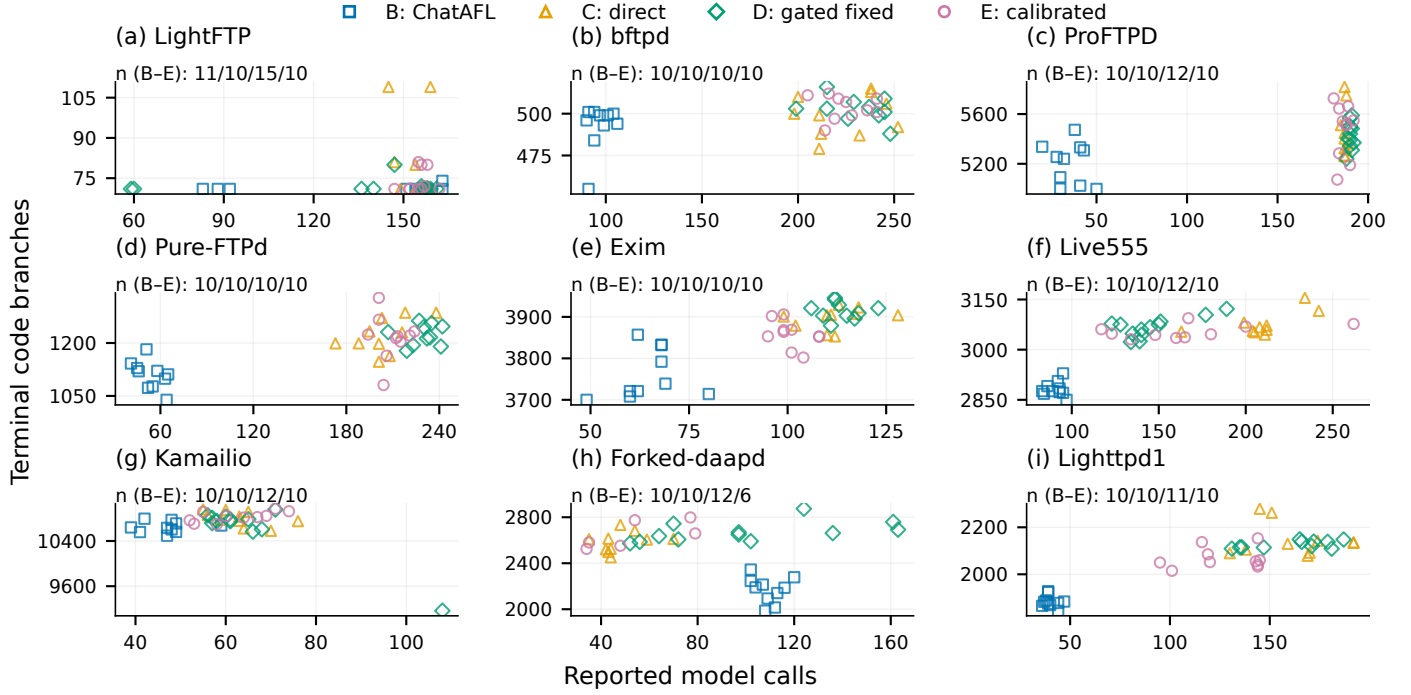


Figure 9: Reported model calls versus terminal code branches, by target and arm.

Table 17: Vulnerability cases and supporting evidence (L: direct observation; R: report).

Case / interface	Reported vulnerable version	Trigger evidence / oracle	Patch or control evidence
CVE-2023-51713 ProFTPD / FTP	Reported vulnerable version; ASan; 65,535-byte command buffer	60,006-byte input; heap-buffer-overflow: one-byte read beyond a 65,568-byte region (L).	Reported security patch; three rejections, no ASan (L); same input/buffer.
CVE-2026-38998 LIVE555 / RTSP	Campaign: 2023.05.10; public report: 2026.02.26 (R)	728-byte public PoC; heap-use-after-free (L). Fuzzer-discovered input absent.	Attempted local patch: 1/1 still triggers (R); upstream-patch comparison absent.
CVE-2026-39863 Kamailio / SIP over TCP	Reported vulnerable version; TCP required	278-byte input; parsed Content-Length -10 (L); overflow predicate, not crash count.	Local overflow guard suppresses diagnostic (R); equivalence to the identified security patch unverified.
SMARTPL availability case forked-daapd / HTTP	Reported version 27.2	148-byte input; lexer errors; health probe fails; process alive (R). Direct replay evidence absent.	No patched replay/CVE; parser reportedly removed in 28.4.

satisfies that condition; neither a CVE identity nor a verified patched counterpart is established.

8.2. Replay outcomes and their denominators

Table 18 preserves each reported batch separately. A numerator counts replay attempts or selected queue inputs satisfying that case’s predicate; the denominator is the number tested. These are not independent 12-hour fuzzer runs, and the main experiment’s nominal $N = 10$ is not substituted for the actual replay denominator. Selected batches, per-input repeats, and later aggregate counts may overlap; they are not pooled into one success rate. In particular, 19/20 selected inputs is not 95% campaign recall, and 8/9 LIVE555 replays is below the prospective 95% stability threshold.

Additional reported crashes and campaigns cannot be linked to the primary benchmark and ablation observations. Crash totals do not count unique vulnerabilities, and durations such as 60 or 1,540 minutes do not specify time to first trigger. Associations among reproducing inputs, verified A–E conditions, model information exposure, and independent root causes are incomplete. The replay outcomes therefore do not establish arm-specific discovery probabilities, novel CVEs, or admission/calibration gains.

8.3. Controlled rediscovery still to be measured

Confirmatory rediscovery requires a vulnerable version and the same version with a minimal security patch, holding dependencies, instrumentation, settings, and input compatibility

Table 18: Reported replay batches and separate controls; original denominators retained.

Case	Vulnerable-side batches (R)	Patched or diagnostic control	Interpretation / missing evidence
CVE-2023-51713	1/1 seed; 1/1 fuzzer crash (Sept. 22); 3/3 batch crashes (Sept. 25).	Reported fix: 0/3 ASan activations; 3/3 FTP 500 rejections (L, Sept. 28).	Separate replay batches; no arm-specific discovery probability or trigger time.
CVE-2026-38998	1/1 public PoC; three fuzzer inputs: 3/3 each; later aggregate: 8/9.	Local patch: 1/1 still triggers (R); successful patched-side denominator absent.	Unpooled batches; fuzzer input/trial records absent; no confirmed patch pair.
CVE-2026-39863	1/1 seed; queue samples: 3/4 (60 s), 19/20 (60 min), 19/20 (1,540 min), 50/51 (separate 60-min batch).	Local guard: fingerprint absent (R); repeat count unavailable.	Queue samples, not campaign recall; no extrapolation to 195 entries; patched-version evidence absent.
SMARTPL availability case	5/5 advisory repeats; 3/3 48-message-seed repeats; 1/1 later advisory replay.	Sampled campaign hangs: 0/5 SMARTPL hits (R); patched replay absent.	Known-input unavailability; no confirmed campaign rediscovery or CVE.

constant. Each case needs a network-reachable trigger, independent reached/triggered criteria, reproducible initial conditions, and a validated patch comparison. The proposed six-case pilot across at least three protocols, followed by expansion toward twelve, remains incomplete. Eligibility requires at least 95% observed oracle repeatability in a declared validation set, no activation after patching, reproducible experimental preparation, and information separation. The four cases do not constitute six qualified version pairs; a small all-success replay batch does not establish an underlying success probability of at least 95%.

Information-controlled campaigns must exclude CVE identifiers, advisories, public PoCs, and patch contents from model inputs and campaign seeds, while retaining anonymous oracle IDs in separate benchmark metadata. The public PoC and known triggers in this collection are validation inputs; their replay is not evidence of information-controlled rediscovery. Source-level trigger conditions, sanitizer reports with patch validation, and protocol invariants distinguish reaching relevant code from actually triggering a defect. Experiments require isolated target networks and reset between trials; these are protocol requirements, not properties inferred from the reports.

For each CVE and arm, the formal plan retains ten independent runs and a 12-hour deadline, prespecified before execution. Deadline recall, time-to-trigger, reached-but-not-triggered counts, patched-target activation, executions/coverage/tokens before trigger, and recall per CPU-hour or 100,000 tokens remain unavailable for the controlled A–E comparison. Nontriggers require right-censoring records, and interrupted runs require separate failure accounting. The available replay batches do not define those risk sets, so Kaplan–Meier estimates or arm-level security-effectiveness claims are not reported.

9. Discussion

9.1. What calibration can and cannot repair

The controller calibrates the instrumental utility of an observed state proxy; it does not recover or verify semantic protocol state. If one response identifier aliases several server states, a single productivity estimate averages them. Reweighting the identifier cannot reconstruct lost distinctions. Richer parsing or

internal-state instrumentation addresses different aspects at the cost of adapter complexity and deployment effort.

Low productivity can arise from a poor response proxy, saturated coverage, restrictive grammar, inadequate seeds, or insufficient energy. Independent reward prevents IPSM novelty from certifying itself but does not automatically diagnose these causes. The boundary analysis includes aliasing, cold start, phase, and target-specific regressions.

A fixed provisional horizon can discard a bridge whose pay-off takes more descendants. Shadow validation and horizon sensitivity estimate this limitation; they do not prove every expired candidate worthless. Similarly, U/R can exclude useful error-handling inputs. Independent oracle retention and false-negative analysis are therefore necessary.

9.2. Cost, nonstationarity, and provider variability

Selection-indexed discounting can retain stale evidence for rarely selected states. Priors and the exploration floor preserve opportunities without guaranteeing fast identification. When code coverage saturates, nearly every reward may be zero, and calibration may offer little benefit.

Model-service nondeterminism persists despite identical prompts and nominal parameters. A shared model name may correspond to different inference versions, repeated requests affect costs, and target responses alter later prompts. Equal-policy/equal-cap comparisons control declared resources; actual usage and cost-matched analysis characterize the remaining tradeoffs. Replay assesses decisions under fixed conditions rather than reproducing an adaptive campaign.

9.3. Evidence required for confirmatory claims

A confirmatory calibration claim requires nondegenerate predictions, reduced investment in branch-unproductive Forked-daapd states when present, and no systematic suppression of `Lighttpd1` solely because of limited IPSM novelty. It also requires acceptable throughput on at least three pilot targets and better predictive diagnostics than the shared prequential base-rate reference. D-condition productivity estimates provide an additional diagnostic; the uncalibrated queue-yield ratio is not a probability. An E–D difference must be supported by a plausible

Table 19: Validity threats, required controls, and evidence limits.

Dimension	Required control	Current limitation
Internal	Admission-only C–D; scheduling-only E–D; repair off; paired initial states.	Missing observations; differing operation order.
Construct	Separate IPSM state edges, code branches, bitmap counts, and reached/triggered events.	Observed proxies incompletely measure the intended criteria.
Statistical	Repeated trials; test families; effect sizes; clustered/hierarchical intervals; censoring.	Partial, unequal-duration data; no confirmatory effects.
Selection	Intention-to-run accounting; prespecified infrastructure repetitions.	Uninitiated trials unobserved; additional records need not be independent.
Reproducibility	Identical versions, conditions, seeds, prompts, and measurement definitions.	Unresolved model versions and incomplete experimental provenance.
External	Multiple targets/protocols; negative cases; adapter assumptions.	Encrypted/opaque binary protocols untested.
Security	Minimal patch pairs; independent triggers; information separation.	Controlled A–E and version-pair evidence incomplete; pretraining leakage unresolved.

measured mechanism. These are prospective eligibility criteria rather than completed significance tests.

When these criteria are unmet, calibration remains exploratory and admission must be assessed independently. Null rediscovery results do not justify selecting easier vulnerabilities after observing outcomes, and unsupported repair remains diagnostic. Parameters and analysis decisions must be fixed before confirmatory observation; subsequent methodological changes require separate evaluation.

The planned confirmatory evidence comprises D–C and E–D estimates, complete candidate-level measurements, at least nine targets, twenty runs per core arm, and a reproducible historical benchmark of approximately eight to twelve vulnerabilities. The four replay cases do not meet that benchmark requirement. These are intended sample sizes and coverage criteria, not achieved results.

10. Threats to Validity

Table 19 summarizes the controls and remaining evidence limits.

The analysis is retrospective. Equivalence between the assessed system and every experimental version is unverified, so methodological concerns and observed outcomes retain separate evidential status. A nonsignificant adjusted comparison does not establish equivalence, and favorable endpoints do not identify a mechanism when duration, resources, or effective procedures differ. Admission, calibration, and rediscovery each require independent supporting evidence.

Benchmark properties can change the relative ranking of fuzzers: initial seed coverage and execution speed are empirically consequential covariates [49]. Consequently, a favorable endpoint under unequal experimental conditions cannot isolate the admission or scheduling effect. Mutation-based assessment offers a complementary fault-oriented perspective [50]; here, code coverage and IPSM growth are kept separate from independently verified vulnerability outcomes.

11. Conclusion

LoopFuzz treats response-derived protocol states and LLM output as uncertain evidence for resource allocation. Its admission criteria separate code progress from IPSM novelty, gives state-only proposals bounded provisional validation, and calibrates state productivity at scheduling episodes. The response-visible interface remains useful, while independent code feedback determines its measured value.

The benchmark and ablation observations describe coverage, model usage, and predictive/admission behavior under incomplete operational validation. Four vulnerability cases provide replay evidence with differing support for patched comparisons. These results do not yet establish causal admission gains, calibration gains, or improved rediscovery. The five-arm protocol identifies the additional measurements required, including target-specific regressions and resource costs.

CRedit authorship contribution statement

Author contribution roles remain to be confirmed by all authors before submission.

Declaration of competing interest

The competing-interest declaration remains to be confirmed by all authors.

Funding

Funding information remains to be confirmed by all authors.

Data availability

The 628 experimental records and four vulnerability case studies are accompanied by trial-level measurements, version and input provenance, and numerical figure data. Public access and persistent identifiers remain unspecified; exact reproduction of model-service responses is not claimed.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex assisted language revision, organization, figure preparation, and consistency assessment. The authors remain responsible for scientific accuracy and the final disclosure.

Appendix A. Complete descriptive results

Table A.1 reports all observations underlying the descriptive comparisons, using the same values and sample SDs as the main results. Sample sizes are shown once per row; unavailable target-arm conditions are not imputed. Row IDs combine arm and target number. Arms A–E retain their main-text meanings; E_{99} and E_{100} denote sensitivity groups at $\gamma = 0.99$ and $\gamma = 1.0$.

1: LightFTP 4: Pure-FTPd 7: Kamailio
2: bftpd 5: Exim 8: Forked-daapd
3: ProFTPD 6: Live555 9: Lighttpd1

AUC: branch-hours over each recorded trajectory. The IPSM column reports state-edge counts. Reported termination categories: C=completion, X=termination near the time limit, S=abort, O=memory exhaustion. Categories do not independently establish causes.

Table A.1: Observed endpoints and full-trajectory AUC (mean $\pm$ SD).

ID	n	Branches	AUC	IPSM Status
A1	13	71.0 $\pm$ 0.0	1758.5 $\pm$ 33.8	187.3 $\pm$ 11.3 C:13
A2	10	486.0 $\pm$ 8.8	11868.7 $\pm$ 165.6	163.1 $\pm$ 15.4 C:10
A3	10	5053.8 $\pm$ 97.4	122652.5 $\pm$ 1730.1	190.5 $\pm$ 27.9 C:10
A4	11	1076.4 $\pm$ 89.6	23054.1 $\pm$ 2084.4	236.5 $\pm$ 26.7 C:11
A5	10	3798.8 $\pm$ 43.3	91083.0 $\pm$ 1031.5	88.8 $\pm$ 5.2 C:10
A6	10	2893.6 $\pm$ 14.5	71127.4 $\pm$ 517.3	96.1 $\pm$ 8.6 C:10
A7	10	10002.6 $\pm$ 144.2	242853.7 $\pm$ 2772.6	109.4 $\pm$ 9.7 C:10
A8	10	2330.8 $\pm$ 60.0	54453.2 $\pm$ 1494.1	21.6 $\pm$ 2.1 C:10
A9	10	1940.7 $\pm$ 21.2	46909.4 $\pm$ 235.7	17.7 $\pm$ 2.5 C:10
B1	11	71.3 $\pm$ 0.9	1728.1 $\pm$ 36.5	195.1 $\pm$ 14.6 C:11
B2	10	492.2 $\pm$ 14.1	12068.6 $\pm$ 323.8	167.1 $\pm$ 12.5 C:10
B3	10	5205.9 $\pm$ 166.7	125217.7 $\pm$ 2989.1	150.0 $\pm$ 18.5 C:10
B4	10	1109.3 $\pm$ 40.0	24976.6 $\pm$ 1126.8	186.0 $\pm$ 31.5 C:10
B5	10	3761.7 $\pm$ 60.4	91301.7 $\pm$ 1501.4	115.8 $\pm$ 13.5 C:10
B6	10	2882.4 $\pm$ 22.0	71224.8 $\pm$ 422.5	143.8 $\pm$ 8.8 C:10
B7	10	10644.3 $\pm$ 96.5	257606.8 $\pm$ 2082.0	106.5 $\pm$ 6.8 C:10
B8	10	2169.6 $\pm$ 113.0	49393.2 $\pm$ 3335.9	17.7 $\pm$ 4.9 C:10
B9	10	1884.1 $\pm$ 25.3	45075.8 $\pm$ 729.3	14.3 $\pm$ 3.1 C:10
C1	10	80.6 $\pm$ 15.5	1987.7 $\pm$ 394.1	210.6 $\pm$ 8.3 C:10
C2	10	498.9 $\pm$ 12.2	12082.0 $\pm$ 451.0	217.9 $\pm$ 13.8 C:6,X:4
C3	10	5446.3 $\pm$ 194.7	130538.5 $\pm$ 4511.6	300.7 $\pm$ 29.1 X:9,C:1
C4	10	1221.2 $\pm$ 48.9	28860.8 $\pm$ 1645.7	328.5 $\pm$ 22.5 C:7,X:3
C5	10	3896.1 $\pm$ 25.6	94880.1 $\pm$ 928.3	150.9 $\pm$ 9.5 C:10
C6	10	3075.7 $\pm$ 34.4	73511.0 $\pm$ 562.9	166.8 $\pm$ 9.3 C:10
C7	10	10800.6 $\pm$ 129.0	260260.9 $\pm$ 2942.4	139.2 $\pm$ 9.8 C:9,X:1
C8	10	2585.4 $\pm$ 86.2	61776.8 $\pm$ 1863.7	22.3 $\pm$ 2.2 C:8,S:2
C9	10	2145.0 $\pm$ 69.8	53003.8 $\pm$ 1613.7	32.1 $\pm$ 5.0 C:10
D1	15	71.7 $\pm$ 2.3	1780.7 $\pm$ 45.2	207.0 $\pm$ 17.0 C:15
D2	10	502.7 $\pm$ 7.5	12267.7 $\pm$ 387.9	226.7 $\pm$ 14.0 C:10
D3	12	5428.1 $\pm$ 104.0	132539.2 $\pm$ 2868.9	305.3 $\pm$ 20.3 C:12
D4	10	1223.3 $\pm$ 29.7	27863.9 $\pm$ 1496.9	321.4 $\pm$ 11.1 C:10
D5	10	3914.9 $\pm$ 20.9	93659.0 $\pm$ 573.1	152.7 $\pm$ 9.3 C:10
D6	12	3067.1 $\pm$ 29.5	75436.1 $\pm$ 712.2	170.1 $\pm$ 11.1 C:12

Table A.1 (continued)

ID	n	Branches	AUC	IPSM Status
D7	12	10631.2 $\pm$ 472.9	254332.1 $\pm$ 9492.1	128.2 $\pm$ 26.6 C:12
D8	12	2670.2 $\pm$ 87.3	63646.7 $\pm$ 1977.0	22.8 $\pm$ 2.9 C:12
D9	11	2126.7 $\pm$ 15.7	51223.2 $\pm$ 621.1	28.9 $\pm$ 2.7 C:11
E1	10	73.9 $\pm$ 4.5	1825.0 $\pm$ 107.6	210.9 $\pm$ 9.3 C:10
E2	10	503.7 $\pm$ 7.1	12416.6 $\pm$ 220.9	215.8 $\pm$ 7.5 C:9,O:1
E3	10	5469.8 $\pm$ 215.7	130564.7 $\pm$ 4604.1	307.8 $\pm$ 21.3 X:5,C:5
E4	10	1215.4 $\pm$ 63.7	27894.3 $\pm$ 1737.6	311.6 $\pm$ 22.4 X:5,C:5
E5	10	3858.3 $\pm$ 32.8	93123.6 $\pm$ 924.6	138.6 $\pm$ 14.3 C:10
E6	10	3054.2 $\pm$ 20.5	72378.5 $\pm$ 1049.8	162.1 $\pm$ 4.4 C:10
E7	10	10830.0 $\pm$ 91.3	259377.2 $\pm$ 2649.6	130.9 $\pm$ 9.0 C:10
E8	6	2648.2 $\pm$ 116.0	62349.5 $\pm$ 1726.9	22.5 $\pm$ 2.1 C:6
E9	10	2068.1 $\pm$ 44.5	50385.1 $\pm$ 941.3	27.1 $\pm$ 4.4 C:10
$E_{99}1$	8	71.0 $\pm$ 0.0	1743.4 $\pm$ 53.5	211.5 $\pm$ 13.7 C:8
$E_{99}2$	20	498.6 $\pm$ 7.6	12314.1 $\pm$ 296.6	223.6 $\pm$ 10.0 C:18,X:2
$E_{99}3$	20	5467.1 $\pm$ 147.2	133069.1 $\pm$ 3936.3	307.4 $\pm$ 20.6 C:19,X:1
$E_{99}4$	20	1200.3 $\pm$ 31.2	27522.1 $\pm$ 1399.2	306.6 $\pm$ 18.2 C:20
$E_{99}5$	20	3846.7 $\pm$ 30.9	94251.8 $\pm$ 941.2	157.4 $\pm$ 12.8 X:3,C:17
$E_{99}6$	3	3117.0 $\pm$ 11.5	74558.9 $\pm$ 497.9	171.3 $\pm$ 9.1 C:3
$E_{99}7$	5	10907.6 $\pm$ 53.8	265303.7 $\pm$ 2194.1	136.2 $\pm$ 5.9 C:4,S:1
$E_{99}9$	12	2069.5 $\pm$ 58.0	50428.9 $\pm$ 1260.2	27.5 $\pm$ 5.3 C:12
$E_{100}1$	4	71.0 $\pm$ 0.0	1726.2 $\pm$ 74.8	215.2 $\pm$ 7.0 C:4
$E_{100}2$	10	500.3 $\pm$ 5.7	12417.2 $\pm$ 192.6	223.1 $\pm$ 10.6 C:9,X:1
$E_{100}3$	10	5429.7 $\pm$ 158.7	131751.6 $\pm$ 3888.9	304.4 $\pm$ 22.5 X:2,C:8
$E_{100}4$	10	1181.2 $\pm$ 23.6	26741.3 $\pm$ 1359.4	302.8 $\pm$ 16.7 X:1,C:9
$E_{100}5$	10	3837.1 $\pm$ 33.1	93785.6 $\pm$ 875.9	155.9 $\pm$ 12.6 C:10
$E_{100}6$	2	3110.5 $\pm$ 3.5	74277.3 $\pm$ 141.8	169.5 $\pm$ 12.0 C:2
$E_{100}7$	3	10909.7 $\pm$ 72.5	264791.4 $\pm$ 2646.5	135.7 $\pm$ 8.1 C:3
$E_{100}9$	6	2065.7 $\pm$ 80.9	50258.1 $\pm$ 1724.9	26.5 $\pm$ 3.1 C:6

References

- [1] V.-T. Pham, M. Böhme, A. Roychoudhury, AFLNet: A greybox fuzzer for network protocols, in: 2020 IEEE 13th International Conference on Software Testing, Verification and Validation (ICST), 2020, pp. 460–465. doi:10.1109/ICST46399.2020.00062.
- [2] R. Natella, StateAFL: Greybox fuzzing for stateful network servers, Empirical Software Engineering 27 (2022). doi:10.1007/s10664-022-10233-3, art. no. 191.
- [3] R. Meng, M. Mirchev, M. Böhme, A. Roychoudhury, Large language model guided protocol fuzzing, in: 31st Annual Network and Distributed System Security Symposium (NDSS 2024), 2024, pp. 1–17. doi:10.14722/ndss.2024.24556.
- [4] S. Qin, F. Hu, Z. Ma, B. Zhao, T. Yin, C. Zhang, NSFuzz: Towards efficient and state-aware network service fuzzing, ACM Transactions on Software Engineering and Methodology 32 (2023) 1–26. doi:10.1145/3580598.
- [5] J. Ba, M. Böhme, Z. Mirzamomen, A. Roychoudhury, Stateful greybox fuzzing, in: 31st USENIX Security Symposium (USENIX Security 2022), 2022, pp. 3255–3272. URL: <https://www.usenix.org/conference/usenixsecurity22/presentation/ba>.
- [6] R. Meng, V.-T. Pham, M. Böhme, A. Roychoudhury, AFLNet Five Years Later: On Coverage-Guided Protocol Fuzzing, IEEE Transactions on Software Engineering 51 (2025) 960–974. doi:10.1109/tse.2025.3535925.
- [7] C. McMahon Stone, S. L. Thomas, M. Vanhoef, J. Henderson, N. Bailluet, T. Chothia, The Closer You Look, The More You Learn: A Grey-box Approach to Protocol State Machine Learning, in: Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security, 2022, pp. 2265–2278. doi:10.1145/3548606.3559365.
- [8] S. Schumilo, C. Aschermann, A. Jemmett, A. Abbasi, T. Holz, Nyx-net: network fuzzing with incremental snapshots, in:

- Proceedings of the Seventeenth European Conference on Computer Systems, 2022, pp. 166–180. doi:10.1145/3492321.3519591.
- [9] A. Andronidis, C. Cadar, SnapFuzz: high-throughput fuzzing of network applications, in: Proceedings of the 31st ACM SIGSOFT International Symposium on Software Testing and Analysis, 2022, pp. 340–351. doi:10.1145/3533767.3534376.
 - [10] N. Bars, M. Schloegel, N. Schiller, L. Bernhard, T. Holz, No Peer, no Cry: Network Application Fuzzing via Fault Injection, in: Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security, 2024, pp. 750–764. doi:10.1145/3658644.3690274.
 - [11] W.-C. Kao, Y.-C. Chen, Y.-S. Lin, Y.-C. Yang, C.-Y. Li, C.-Y. Huang, BLuEMan: A stateful simulation-based fuzzing framework for Open-Source RTOS bluetooth low energy protocol stacks, in: 34th USENIX Security Symposium (USENIX Security 25), USENIX Association, Seattle, WA, 2025, pp. 6259–6278. URL: <https://www.usenix.org/conference/usenixsecurity25/presentation/kao>.
 - [12] Y. Zou, J.-J. Bai, Z.-M. Jiang, M. Zhao, D. Zhou, Blackbox Fuzzing of Distributed Systems with Multi-Dimensional Inputs and Symmetry-Based Feedback Pruning, in: Proceedings 2025 Network and Distributed System Security Symposium, 2025. doi:10.14722/ndss.2025.241912.
 - [13] P. Srivastava, M. Payer, Gramatron: effective grammar-aware fuzzing, in: Proceedings of the 30th ACM SIGSOFT International Symposium on Software Testing and Analysis, 2021, pp. 244–256. doi:10.1145/3460319.3464814.
 - [14] D. Wang, Y. Li, Z. Zhang, K. Chen, CarpetFuzz: Automatic program option constraint extraction from documentation for fuzzing, in: 32nd USENIX Security Symposium (USENIX Security 2023), 2023, pp. 1919–1936. URL: <https://www.usenix.org/conference/usenixsecurity23/presentation/wang-dawei>.
 - [15] R. Dutra, R. Gopinath, A. Zeller, FormatFuzzer: Effective Fuzzing of Binary File Formats, ACM Transactions on Software Engineering and Methodology 33 (2024) 1–29. doi:10.1145/3628157.
 - [16] N. Bars, M. Schloegel, T. Scharnowski, N. Schiller, T. Holz, Fuzztruction: Using fault injection-based fuzzing to leverage implicit domain knowledge, in: 32nd USENIX Security Symposium (USENIX Security 23), USENIX Association, Anaheim, CA, 2023, pp. 1847–1864. URL: <https://www.usenix.org/conference/usenixsecurity23/presentation/bars>.
 - [17] C. Lyu, J. Xu, S. Ji, X. Zhang, Q. Wang, B. Zhao, G. Pan, W. Cao, P. Chen, R. Beyah, MINER: A hybrid Data-Driven approach for REST API fuzzing, in: 32nd USENIX Security Symposium (USENIX Security 23), USENIX Association, Anaheim, CA, 2023, pp. 4517–4534. URL: <https://www.usenix.org/conference/usenixsecurity23/presentation/lyu>.
 - [18] J. Wang, L. Yu, X. Luo, LLMIF: Augmented large language model for fuzzing IoT devices, in: 2024 IEEE Symposium on Security and Privacy (SP), 2024, pp. 881–896. doi:10.1109/sp54263.2024.00211.
 - [19] C. S. Xia, M. Paltenghi, J. L. Tian, M. Pradel, L. Zhang, Fuzz4All: Universal fuzzing with large language models, in: Proceedings of the IEEE/ACM 46th International Conference on Software Engineering, 2024, pp. 1–13. doi:10.1145/3597503.3639121.
 - [20] Y. Deng, C. S. Xia, C. Yang, S. D. Zhang, S. Yang, L. Zhang, Large Language Models are Edge-Case Generators: Crafting Unusual Programs for Fuzzing Deep Learning Libraries, in: Proceedings of the IEEE/ACM 46th International Conference on Software Engineering, 2024, pp. 1–13. doi:10.1145/3597503.3623343.
 - [21] C. Yang, Y. Deng, R. Lu, J. Yao, J. Liu, R. Jabbarvand, L. Zhang, WhiteFox: White-Box Compiler Fuzzing Empowered by Large Language Models, Proceedings of the ACM on Programming Languages 8 (2024) 709–735. doi:10.1145/3689736.
 - [22] D. Wang, G. Zhou, L. Chen, D. Li, Y. Miao, ProphetFuzz: Fully Automated Prediction and Fuzzing of High-Risk Option Combinations with Only Documentation via Large Language Model, in: Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security, 2024, pp. 735–749. doi:10.1145/3658644.3690231.
 - [23] R. Meng, G. J. Duck, A. Roychoudhury, Large Language Model Assisted Hybrid Fuzzing, IEEE Transactions on Software Engineering 52 (2026) 2367–2382. doi:10.1109/tse.2026.3694408.
 - [24] H. Xu, Y. Zhao, J. Yu, H. Wang, HGFuzzer: Directed Greybox Fuzzing via Large Language Model, ACM Transactions on Software Engineering and Methodology (2026). doi:10.1145/3841476, advance online publication, August 22, 2026.
 - [25] A. Borcharding, M. Giraud, I. Fitzgerald, J. Beyerer, The bandit’s states: Modeling state selection for stateful network fuzzing as multi-armed bandit problem, in: 2023 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW), 2023, pp. 345–350. doi:10.1109/EuroSPW59978.2023.00043.
 - [26] W. Zhang, J. Chen, S. Cai, K. Wang, Y. Liu, H. Ding, A novel seed scheduling scheme using Thompson sampling for coverage-guided greybox fuzzing, Journal of Systems and Software 236 (2026) 112794. doi:10.1016/j.jss.2026.112794.
 - [27] K. Jian, Y. Zou, C. Wang, N. Li, M. Li, W. Huo, State significance-guided fuzzing for stateful protocol program, in: P. Rümmer, Z. Wu (Eds.), Theoretical Aspects of Software Engineering, volume 15841 of *Lecture Notes in Computer Science*, Springer Nature Switzerland, Cham, 2026, pp. 361–379. doi:10.1007/978-3-031-98208-8_21.
 - [28] D. She, A. Storek, Y. Xie, S. Kweon, P. Srivastava, S. Jana, FOX: Coverage-guided Fuzzing as Online Stochastic Control, in: Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security, 2024, pp. 765–779. doi:10.1145/3658644.3670362.
 - [29] H. Zheng, J. Zhang, Y. Huang, Z. Ren, H. Wang, C. Cao, Y. Zhang, F. Toffalini, M. Payer, FISHFUZZ: Catch deeper bugs by throwing larger nets, in: 32nd USENIX Security Symposium (USENIX Security 23), USENIX Association, Anaheim, CA, 2023, pp. 1343–1360. URL: <https://www.usenix.org/conference/usenixsecurity23/presentation/zheng>.
 - [30] X. Zhu, E. Dai, X. Feng, S. Wang, X. Xia, S. Wen, K.-Y. Lam, Y. Xiang, WingMuzz: Blackbox Testing of IoT Protocols via Two-dimensional Fuzzing Schedule, in: 2025 40th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2025, pp. 2580–2592. doi:10.1109/ase63991.2025.00212.
 - [31] X. Song, L. Pei, J. Wu, Y. Zeng, G. He, C. Zuo, X. Liu, Q. Zhao, S. Guo, X. Liu, ProtocolGuard: Detecting Protocol Non-compliance Bugs via LLM-guided Static Analysis and Dynamic Verification, in: Proceedings 2026 Network and Distributed System Security Symposium, 2026. doi:10.14722/ndss.2026.240521.
 - [32] T. Yang, Y. Qin, L. Zhang, Z. Fu, J. Chen, J. Wang, S. Zhao, Q. Li, R. Li, H. Wang, Y. Zhang, BSFuzzer: Context-Aware Semantic Fuzzing for BLE Logic Flaw Detection, in: Proceedings 2026 Network and Distributed System Security Symposium, 2026. doi:10.14722/ndss.2026.240094.
 - [33] K. Wang, J. Chen, P. Chen, J. Zhuge, J. Bai, H. Duan, Identifying Logical Vulnerabilities in QUIC Implementations, in: Proceedings 2026 Network and Distributed System Security Symposium, 2026. doi:10.14722/ndss.2026.231777.
 - [34] A. Hazimeh, A. Herrera, M. Payer, Magma: A ground-truth fuzzing benchmark, Proceedings of the ACM on Measurement and Analysis of Computing Systems 4 (2020) 1–29. doi:10.1145/3428334.
 - [35] R. Natella, V.-T. Pham, ProFuzzBench: A benchmark for stateful protocol fuzzing, in: Proceedings of the 30th ACM SIGSOFT International Symposium on Software Testing and Analysis, 2021, pp. 662–665. doi:10.1145/3460319.3469077.

- [36] J. Metzman, L. Szekeres, L. Simon, R. Sprabery, A. Arya, FuzzBench: an open fuzzer benchmarking platform and service, in: Proceedings of the 29th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, 2021, pp. 1393–1403. doi:10.1145/3468264.3473932.
- [37] M. Böhme, L. Szekeres, J. Metzman, On the reliability of coverage-based fuzzer benchmarking, in: Proceedings of the 44th International Conference on Software Engineering, 2022, pp. 1621–1633. doi:10.1145/3510003.3510230.
- [38] M. Schloegel, N. Bars, N. Schiller, L. Bernhard, T. Scharnowski, A. Crump, A. Ale-Ebrahim, N. Bissantz, M. Muench, T. Holz, SoK: Prudent evaluation practices for fuzzing, in: 2024 IEEE Symposium on Security and Privacy (SP), 2024, pp. 1974–1993. doi:10.1109/sp54263.2024.00137.
- [39] W. R. Thompson, On the likelihood that one unknown probability exceeds another in view of the evidence of two samples, *Biometrika* 25 (1933) 285–294. doi:10.1093/biomet/25.3-4.285.
- [40] J. Ounjai, V. Wüstholtz, M. Christakis, Green Fuzzer Benchmarking, in: Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis, 2023, pp. 1396–1406. doi:10.1145/3597926.3598144.
- [41] G. W. Brier, Verification of forecasts expressed in terms of probability, *Monthly Weather Review* 78 (1950) 1–3. doi:10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
- [42] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning, volume 70 of *Proceedings of Machine Learning Research*, PMLR, 2017, pp. 1321–1330. URL: <https://proceedings.mlr.press/v70/guo17a.html>.
- [43] J. Davis, M. Goadrich, The relationship between Precision-Recall and ROC curves, in: Proceedings of the 23rd International Conference on Machine Learning, 2006, pp. 233–240. doi:10.1145/1143844.1143874.
- [44] B. Efron, Bootstrap methods: Another look at the jackknife, *The Annals of Statistics* 7 (1979). doi:10.1214/aos/1176344552.
- [45] H. B. Mann, D. R. Whitney, On a test of whether one of two random variables is stochastically larger than the other, *The Annals of Mathematical Statistics* 18 (1947) 50–60. doi:10.1214/aoms/1177730491.
- [46] N. Cliff, Dominance statistics: Ordinal analyses to answer ordinal questions, *Psychological Bulletin* 114 (1993) 494–509. doi:10.1037/0033-2909.114.3.494.
- [47] Y. Benjamini, Y. Hochberg, Controlling the false discovery rate: A practical and powerful approach to multiple testing, *Journal of the Royal Statistical Society Series B: Statistical Methodology* 57 (1995) 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.
- [48] E. L. Kaplan, P. Meier, Nonparametric estimation from incomplete observations, *Journal of the American Statistical Association* 53 (1958) 457–481. doi:10.1080/01621459.1958.10501452.
- [49] D. Wolff, M. Böhme, A. Roychoudhury, Fuzzing: On Benchmarking Outcome as a Function of Benchmark Properties, *ACM Transactions on Software Engineering and Methodology* 35 (2026) 1–26. doi:10.1145/3732936.
- [50] P. Görz, B. Mathis, K. Hassler, E. Güler, T. Holz, A. Zeller, R. Gopinath, Systematic assessment of fuzzers using mutation analysis, in: 32nd USENIX Security Symposium (USENIX Security 23), USENIX Association, Anaheim, CA, 2023, pp. 4535–4552. URL: <https://www.usenix.org/conference/usenixsecurity23/presentation/gorz>.